Course 6: Present and Future of AI



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Present and future of AI

The present

- Assistants, reasoning models, **agents** acting on a computer,
- Models that generate **music**, images, video,
- Measured costs: **electricity**, money, dependence.

The future

- Not a prediction: **choices** being made now,
- By companies, states, artists… and engineers,
- Three open questions, each with **two sides**.

1. Work
Will AI destroy jobs?

2. Environment or sovereignty
Should France build its own frontier model?

3. Artistic expression
Generative AI: Instrument, or parasite of the artists?

The format

Each theme: a few slides of **evidence**, then **one** question with two sides. The next two sessions: you prepare, then you **debate**.

Outline

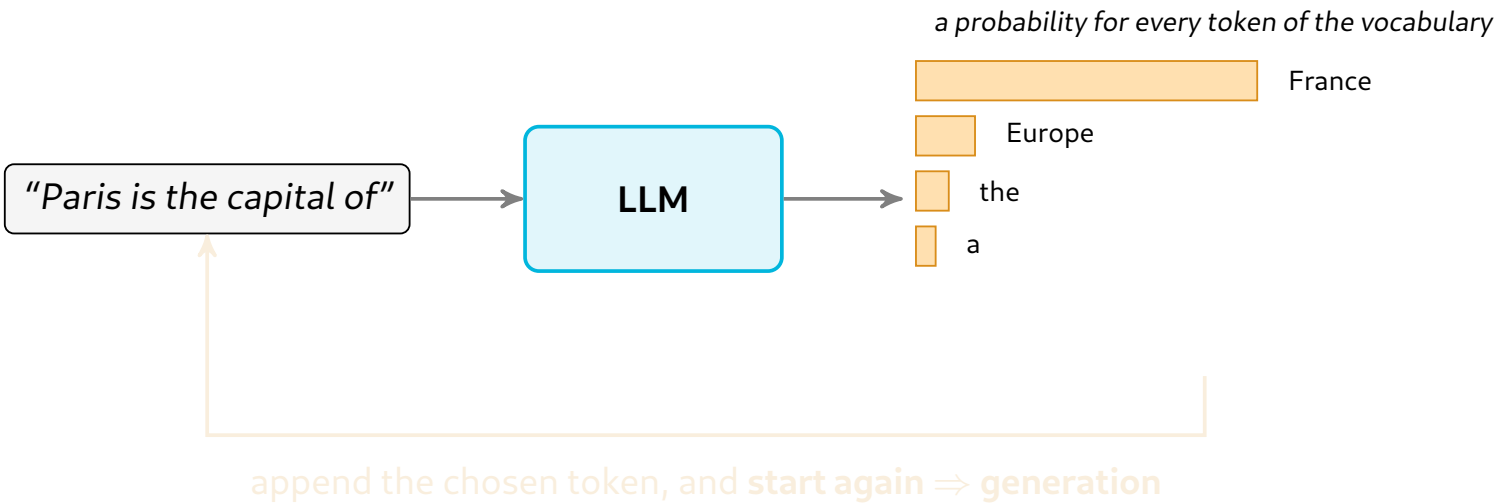
1 Agentic AI: Will AI destroy jobs?

2 Environment or sovereignty: should France build its own model?

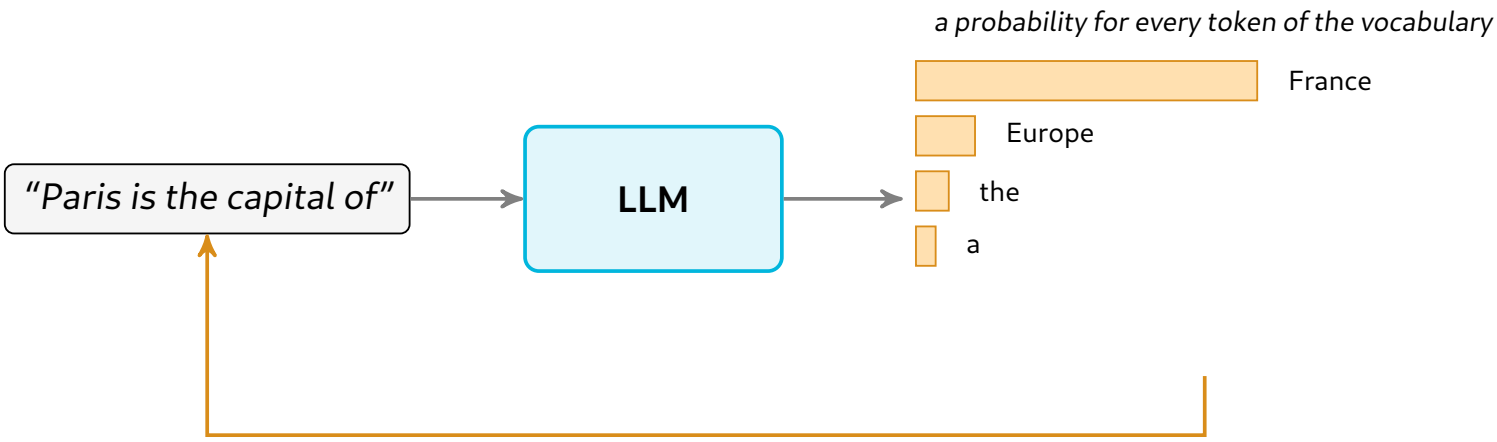
3 AI in artistic expression

4 Next sessions: the debate

Using a Large Language Model (LLM) to generate text



Using a Large Language Model (LLM) to generate text



From text completion to instruction following

The problem

Pre-training teaches the model to **continue text**, not to **answer**.

Before (raw pre-trained)

You: What is the capital of France?

Model: What is the capital of Germany? What is the capital of Spain? ...

it just keeps the pattern going

After (instruction-tuned)

You: What is the capital of France?

Model: Paris.

it answers the actual question

Base model

continues text

instruction tuning

supervised learning

on (instruction → answer) pairs

Assistant

follows instructions

From text completion to instruction following

The problem

Pre-training teaches the model to **continue text**, not to **answer**.

Before (raw pre-trained)

You: What is the capital of France?

Model: What is the capital of Germany? What is the capital of Spain? ...

it just keeps the pattern going

After (instruction-tuned)

You: What is the capital of France?

Model: Paris.

it answers the actual question

Base model

continues text

instruction tuning

supervised learning
on (instruction → answer) pairs

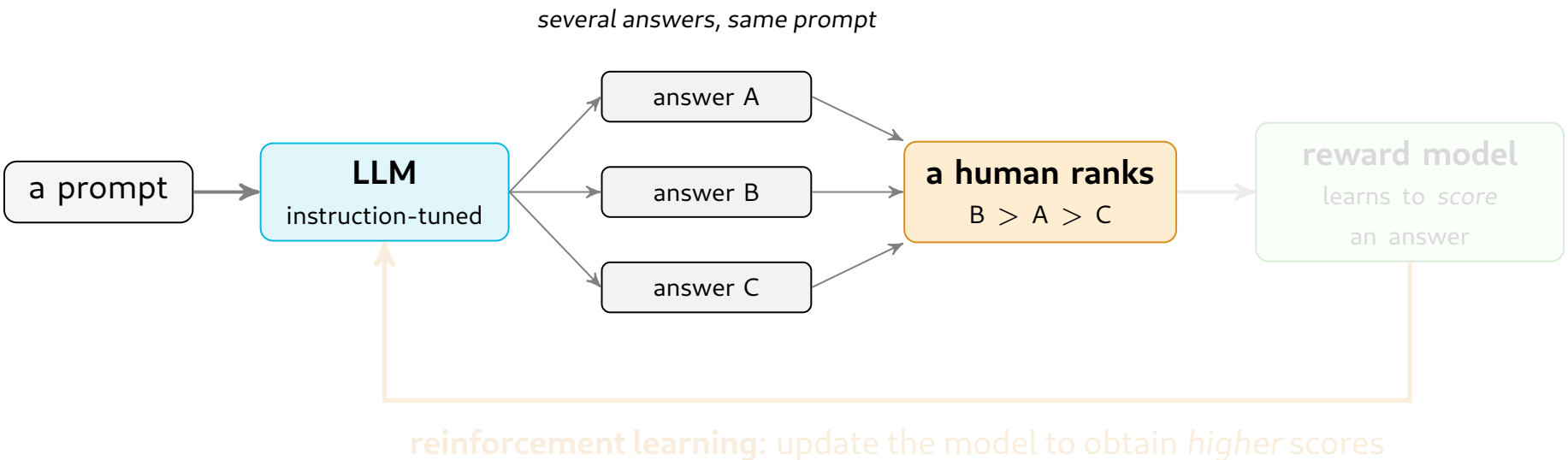
Assistant

follows instructions

Reinforcement Learning from Human Feedback

RLHF: Put human in the loop

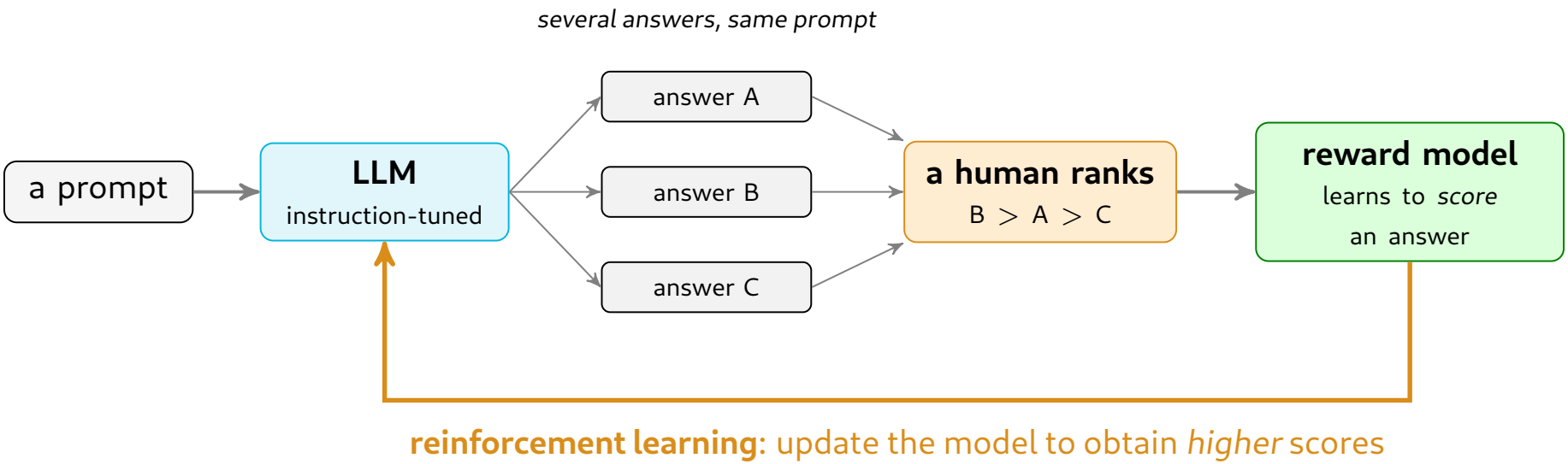
Using human evaluation to reward the model for producing better answers, alignes with pre-defined criteria (values, ethics, safety, ...).



Reinforcement Learning from Human Feedback

RLHF: Put human in the loop

Using human evaluation to reward the model for producing better answers, alignes with pre-defined criteria (values, ethics, safety, ...).



Assistants are not just LLMs



RLHF and sycophancy

The model optimizes **what humans approve of**, based on pre-defined criteria.
Hence the behavior of some assistants: sympathic, enthusiastic, ... but sometimes wrong to please you ("sycophancy").

RLHF is subject to the will of people training it: they may also be used to make a model antipathic, aggressive, manipulative, or else!

Reasoning models: Chain-of-Thought

Giving time to think

Taking time to write a reasoning process before answering.
Leads to better answers on arithmetic, logic, and planning problems, but increases the computation cost.

Direct answer

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 43 ×

it guesses in one shot

Chain-of-thought

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 6 cans × 7 = 42;
5 + 42 = **47**. ✓

the steps first, then the answer

Chain-of-Thought largely participates in the recent progesses of LLMs!

Reasoning models: Chain-of-Thought

Giving time to think

Taking time to write a reasoning process before answering.
Leads to better answers on arithmetic, logic, and planning problems, but increases the computation cost.

Direct answer

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 43 ×

it guesses in one shot

Chain-of-thought

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 6 cans $\times$ 7 $=$ 42;
5 $+$ 42 $=$ **47**. ✓

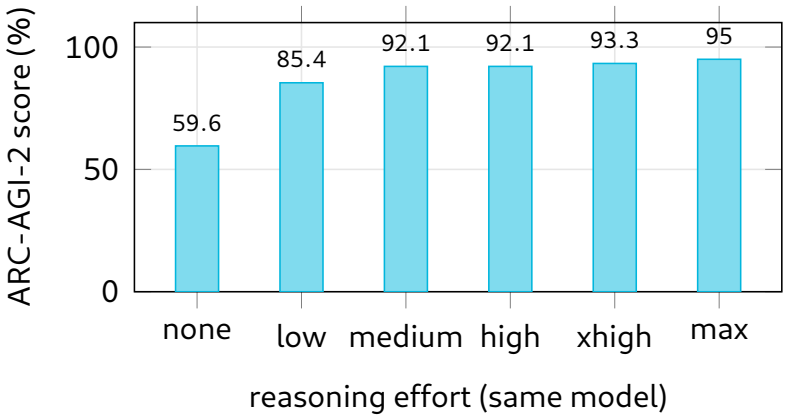
the steps first, then the answer

Chain-of-Thought largely participates in the recent progesses of LLMs!

CoT performance, and its cost

Two places to spend compute

Training: paid once, the model improves for every user.
Inference (CoT): paid at **every question**, and you choose how much.



Longer reasoning improves accuracy!

GPT-6 Astra: **+35 points**, no retraining.

It is also costly

- Paid at **every query**, plus **latency** and **energy**,
- Prices reduce with time (87.5% on ARC-AGI-1 cost **$4,560**/task (2024), **$0.30** (2026))...
- ... but increasing the performance requires larger CoT: +26 points (none → low), then only +10 (low → max) for 100× more compute.
- The frontier stays expensive: one ARC-AGI-3 run with GPT6-Astra costs **$26,098**.

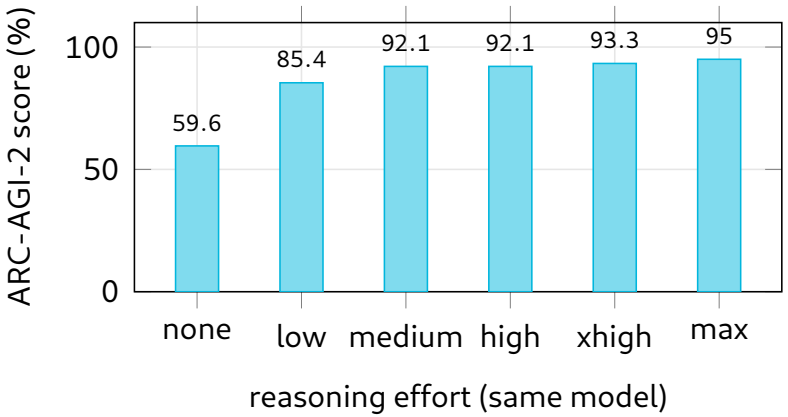
Reasoning is not free: its cost moved from training to every answer.

ARC Prize Verified: GPT-6 Astra, Sep 2026, `https://arcprize.org/results/openai-gpt-6-astra` Grok 4.6 ($0.30), Aug 2026, `https://arcprize.org/results/xai-grok-4-6` o3 ($4,560), Dec 2024, `https://arcprize.org/blog/oai-o3-pub-breakthrough`

CoT performance, and its cost

Two places to spend compute

Training: paid once, the model improves for every user.
Inference (CoT): paid at **every question**, and you choose how much.



Longer reasoning improves accuracy!

GPT-6 Astra: **+35 points**, no retraining.

It is also costly

- Paid at **every query**, plus **latency** and **energy**,
- Prices reduce with time (87.5% on ARC-AGI-1 cost **$4,560**/task (2024), **$0.30** (2026))...
- ... but increasing the performance requires larger CoT: +26 points (none → low), then only +10 (low → max) for 100× more compute.
- The frontier stays expensive: one ARC-AGI-3 run with GPT6-Astra costs **$26,098**.

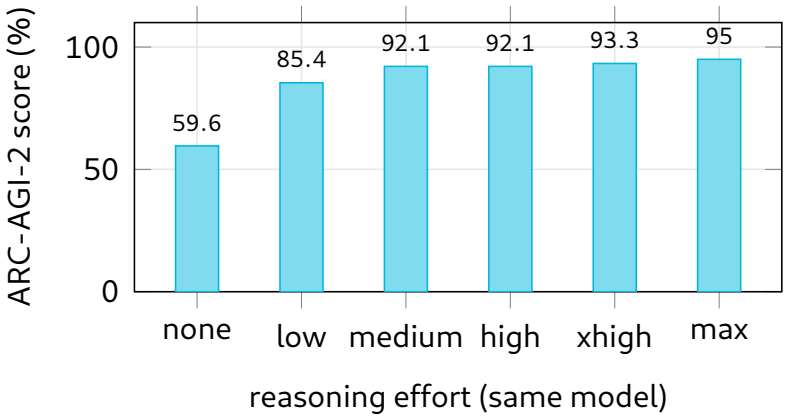
Reasoning is not free: its cost moved from training to every answer.

ARC Prize Verified: GPT-6 Astra, Sep 2026, `https://arcprize.org/results/openai-gpt-6-astra` Grok 4.6 ($0.30), Aug 2026, `https://arcprize.org/results/xai-grok-4-6` o3 ($4,560), Dec 2024, `https://arcprize.org/blog/oai-o3-pub-breakthrough`

CoT performance, and its cost

Two places to spend compute

Training: paid once, the model improves for every user.
Inference (CoT): paid at **every question**, and you choose how much.



Longer reasoning improves accuracy!

GPT-6 Astra: **+35 points**, no retraining.

It is also costly

- Paid at **every query**, plus **latency** and **energy**,
- Prices reduce with time (87.5% on ARC-AGI-1 cost **$4,560**/task (2024), **$0.30** (2026))...
- ... but increasing the performance requires larger CoT: +26 points (none → low), then only +10 (low → max) for 100× more compute.
- The frontier stays expensive: one ARC-AGI-3 run with GPT6-Astra costs **$26,098**.

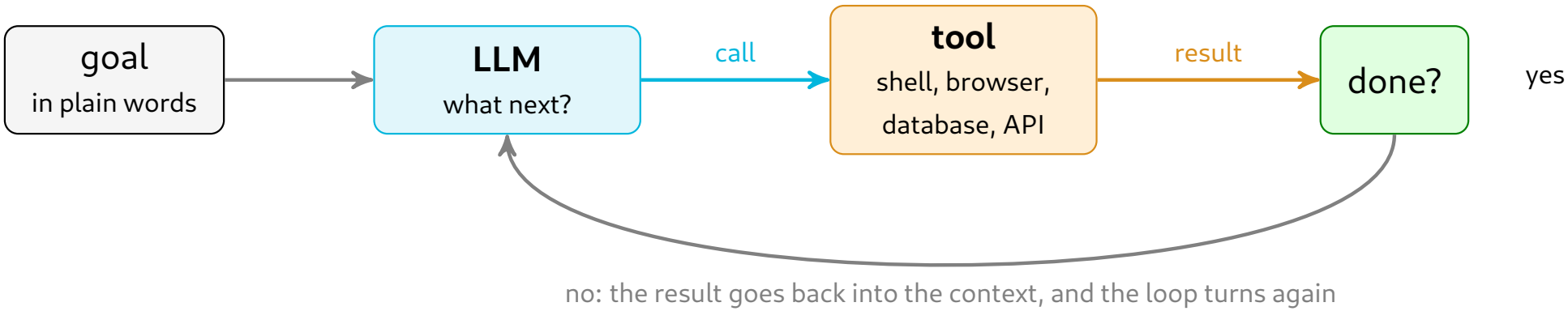
Reasoning is not free: its cost moved from training to every answer.

ARC Prize Verified: GPT-6 Astra, Sep 2026, `https://arcprize.org/results/openai-gpt-6-astra` Grok 4.6 ($0.30), Aug 2026, `https://arcprize.org/results/xai-grok-4-6` o3 ($4,560), Dec 2024, `https://arcprize.org/blog/oai-o3-pub-breakthrough`

Agentic AI: from answers to actions

Agents

An **agent** is an LLM that can call tools, observe the results, and decide what to do next, in a loop, until the goal is reached.

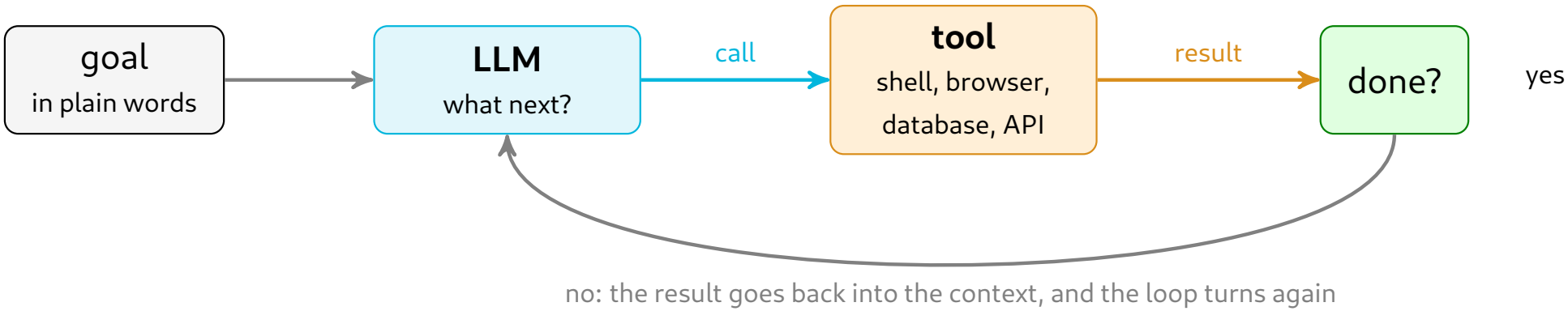


Nothing was added to the model. It was given a loop, and permission.
The model is now able to take actions!

Agentic AI: from answers to actions

Agents

An **agent** is an LLM that can call tools, observe the results, and decide what to do next, in a loop, until the goal is reached.



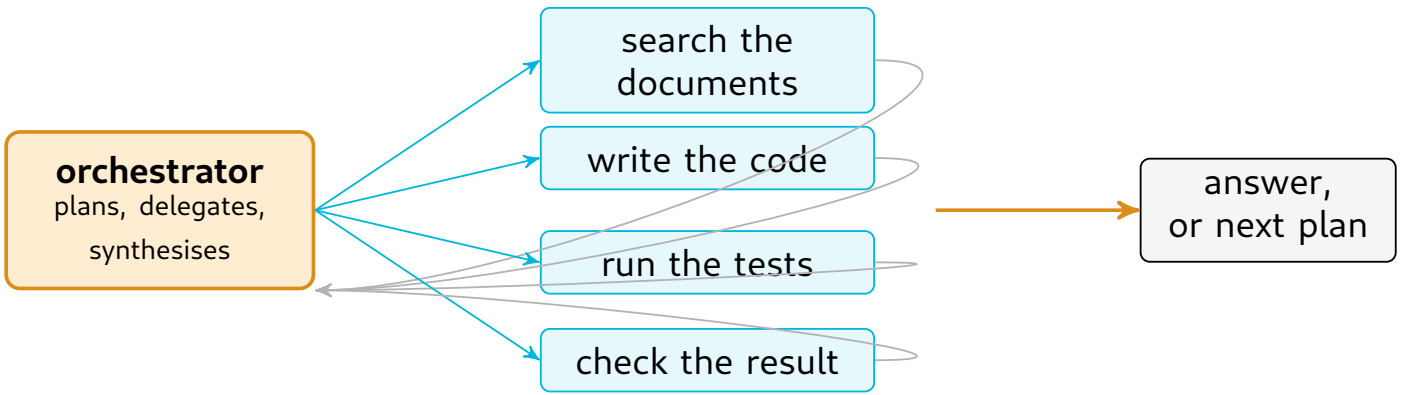
Nothing was added to the model. It was given a loop, and permission.
The model is now able to take actions!

The LLM is an orchestrator of many agents

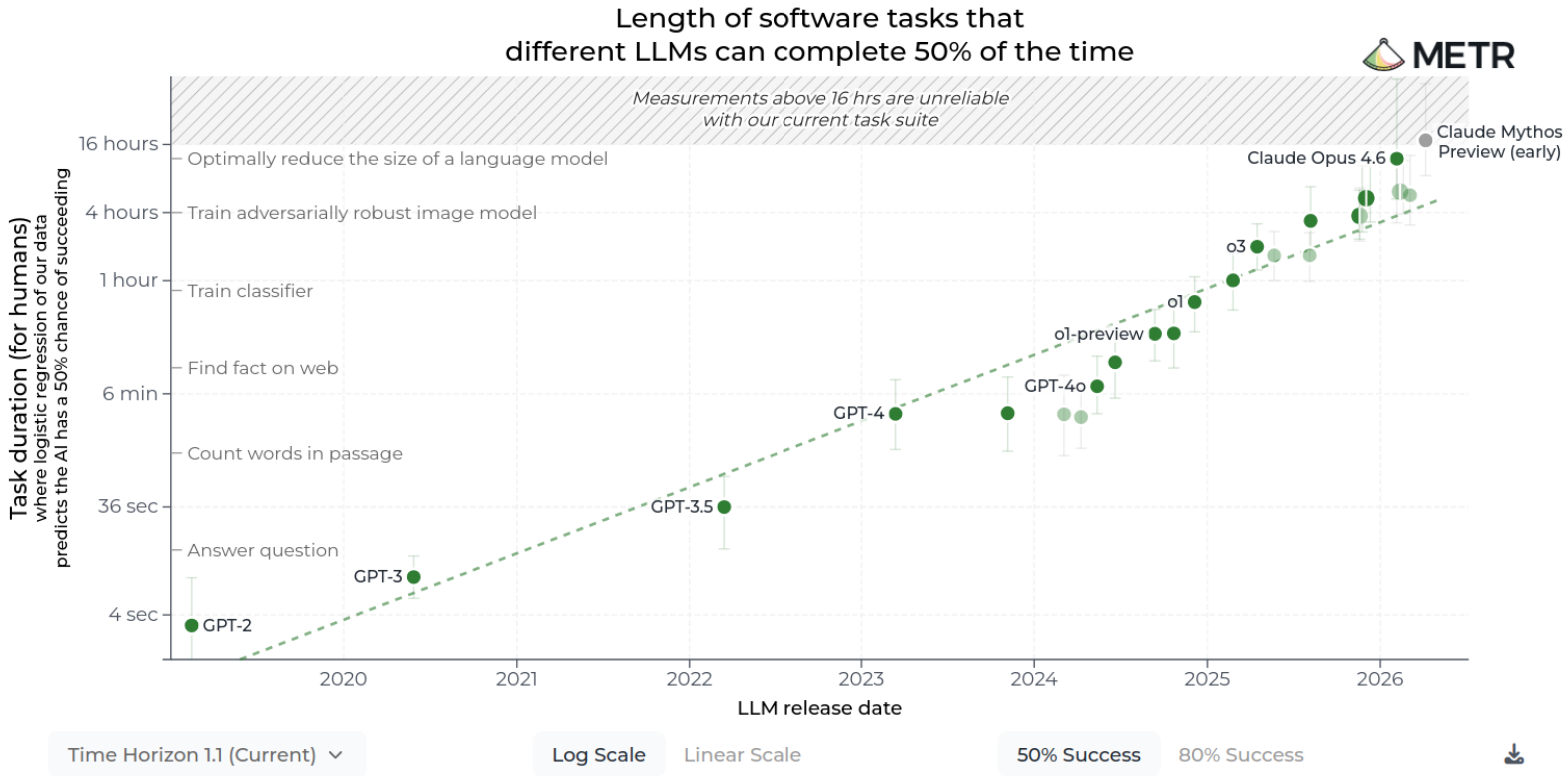
The LLM orchestrates subtasks, but does not take action itself.

1. The LLM understands the overall task and breaks it down into smaller, manageable subtasks.
2. It **delegates** the pieces (to other model instances, or to deterministic tools).
3. It **recombines** the results, and decides whether to continue or to return the final answer.

Notice that the subtasks are decided at run time, not written in advance.



Agents can perform longer and longer tasks



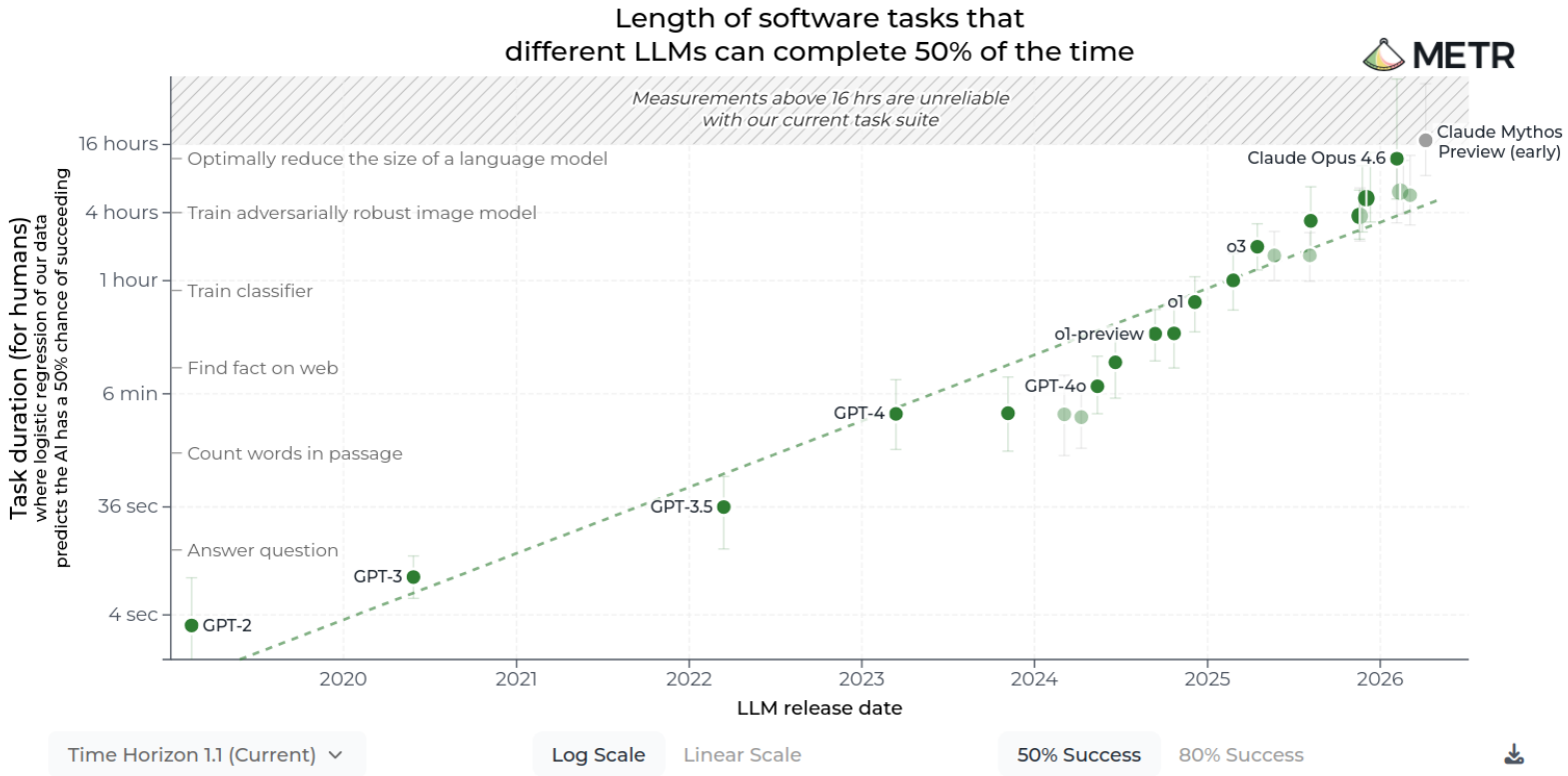
Read it as time saved using agents

2023: a few minutes of an engineer's work. / 2026: a **working day**, and more.
But 50% is not reliable: **someone still has to check**.

Models are progressing very fast $\Longrightarrow$ more and more useful.

METR, *Measuring AI Ability to Complete Long Tasks*, `https://arxiv.org/abs/2503.14499` Time Horizon 1.1, figure from `https://metr.org/time-horizons/` (Oct 2026)

Agents can perform longer and longer tasks



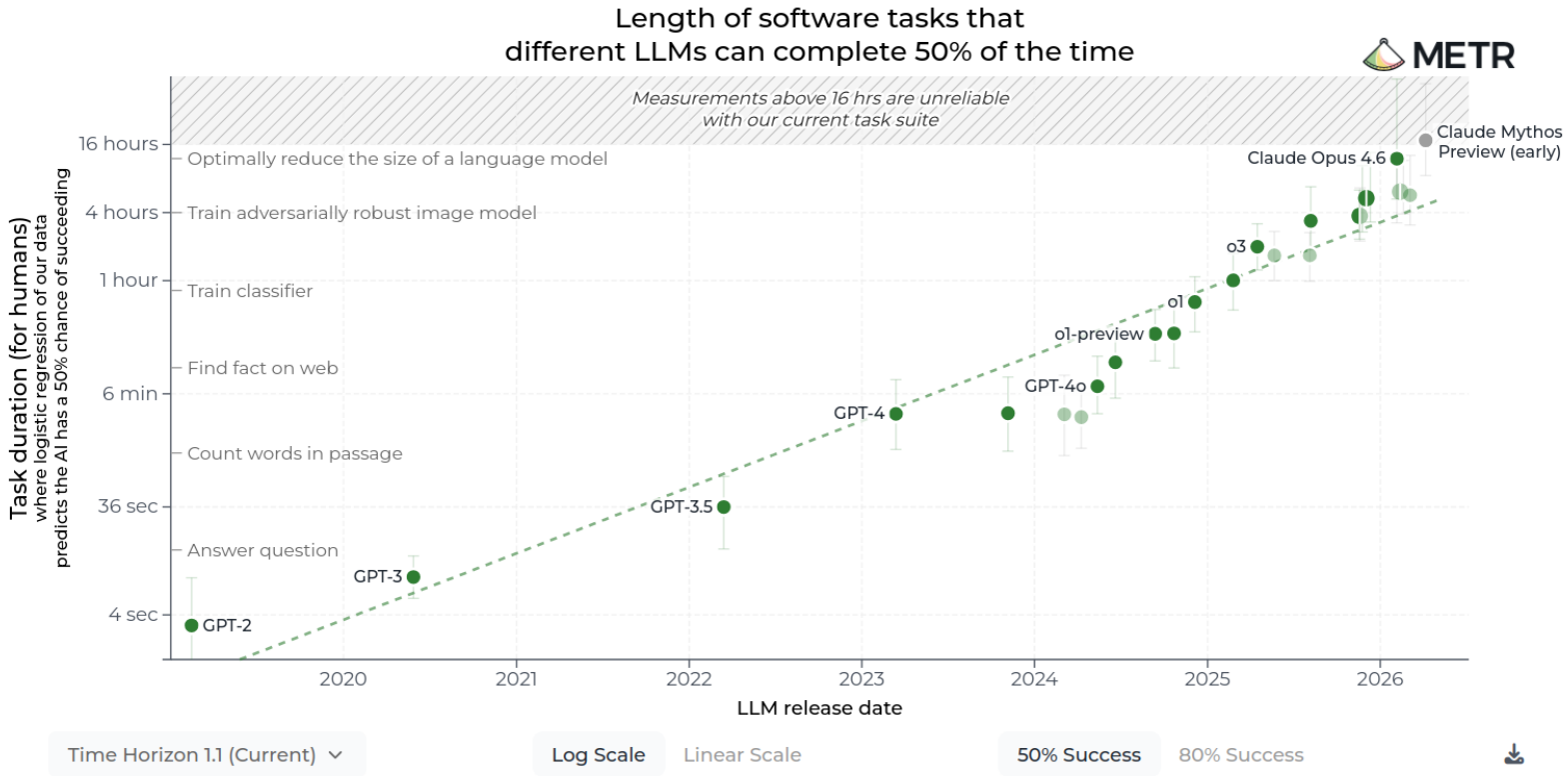
Read it as time saved using agents

2023: a few minutes of an engineer's work. / 2026: a **working day**, and more.
But 50% is not reliable: **someone still has to check**.

Models are progressing very fast $\Longrightarrow$ more and more useful.

METR, *Measuring AI Ability to Complete Long Tasks*, `https://arxiv.org/abs/2503.14499` Time Horizon 1.1, figure from `https://metr.org/time-horizons/` (Oct 2026)

Agents can perform longer and longer tasks



Read it as time saved using agents

2023: a few minutes of an engineer's work. / 2026: a **working day**, and more.
But 50% is not reliable: **someone still has to check**.

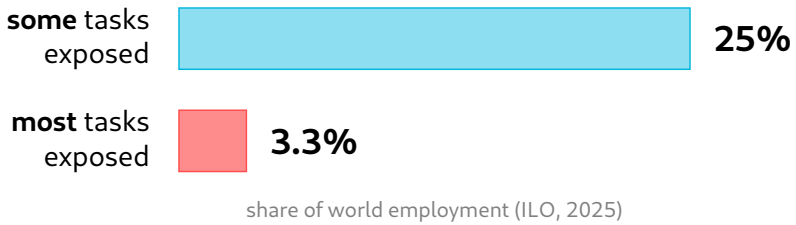
Models are progressing very fast $\Longrightarrow$ more and more useful.

METR, *Measuring AI Ability to Complete Long Tasks*, `https://arxiv.org/abs/2503.14499` Time Horizon 1.1, figure from `https://metr.org/time-horizons/` (Oct 2026)

What is the expected impact of AI on jobs?

1. Exposure

Which tasks of a job can AI do?



2. Productivity

How much faster with AI? (randomised trials)

+26% tasks done
developers, three firms

−40% time spent
professional writing

gains **largest for juniors**

Expected impact: AI transforms jobs more than it replaces them.

But: young people are hired less

US, most exposed jobs, employment vs. expected

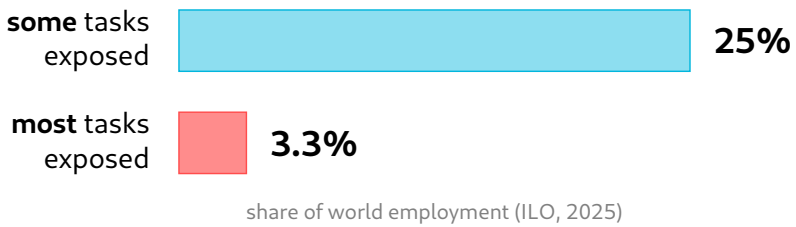
fewer **hirings**, not layoffs

ILO Working Paper 140, May 2025, `https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf` Cui et al., *Management Science*, 2025 Noy & Zhang, *Science*, 2023 Brynjolfsson et al., Aug 2026 (more in the notes)

What is the expected impact of AI on jobs?

1. Exposure

Which tasks of a job can AI do?



2. Productivity

How much faster with AI? (randomised trials)

+26% tasks done
developers, three firms

−40% time spent
professional writing

gains **largest for juniors**

Expected impact: AI transforms jobs more than it replaces them.

But: young people are hired less

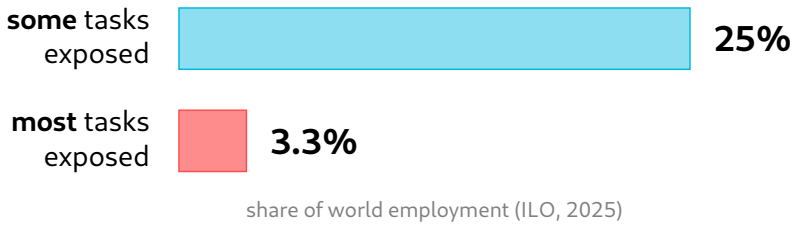
US, most exposed jobs, employment vs. expected

ILO Working Paper 140, May 2025, `https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf` Cui et al., *Management Science*, 2025 Noy & Zhang, *Science*, 2023 Brynjolfsson et al., Aug 2026 (more in the notes)

What is the expected impact of AI on jobs?

1. Exposure

Which tasks of a job can AI do?



2. Productivity

How much faster with AI? (randomised trials)

+26% tasks done
developers, three firms

−40% time spent
professional writing

gains **largest for juniors**

Expected impact: AI transforms jobs more than it replaces them.

But: young people are hired less

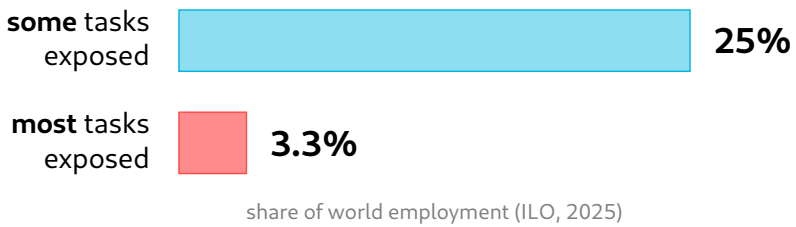
US, most exposed jobs, employment vs. expected

ILO Working Paper 140, May 2025, `https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf` Cui et al., *Management Science*, 2025 Noy & Zhang, *Science*, 2023 Brynjolfsson et al., Aug 2026 (more in the notes)

What is the expected impact of AI on jobs?

1. Exposure

Which tasks of a job can AI do?



2. Productivity

How much faster with AI? (randomised trials)

+26% tasks done
developers, three firms

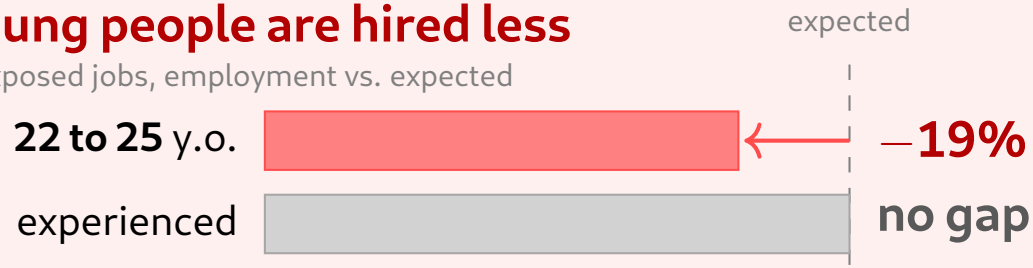
−40% time spent
professional writing

gains **largest for juniors**

Expected impact: AI transforms jobs more than it replaces them.

But: young people are hired less

US, most exposed jobs, employment vs. expected



fewer **hirings**, not layoffs

ILO Working Paper 140, May 2025, `https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf` Cui et al., *Management Science*, 2025 Noy & Zhang, *Science*, 2023 Brynjolfsson et al., Aug 2026 (more in the notes)

Place yourself: will AI destroy jobs?

pick one side

YES *or* **NO**

Strongest for "yes"

- Task length doubles every $\approx$ **4 months** (12 h in 2026),
- Each answer has a **price per task**, falling fast (\$4,560 $\rightarrow$ \$0.30): compare it with a wage,
- Young people already hired less (-19%),
- No juniors today $\Longrightarrow$ no seniors tomorrow.

Strongest for "no"

- "Exposed" $\neq$ replaced: **3.3%** of jobs have most tasks exposed,
- Measured gains: **same person, faster** (+26% tasks),
- 50% success only: **someone still has to check**,
- 60% of 2018 jobs did not exist in 1940.

Autor et al., *QJE*, 2024 (jobs of 1940 and 2018) other sources: previous slides

Outline

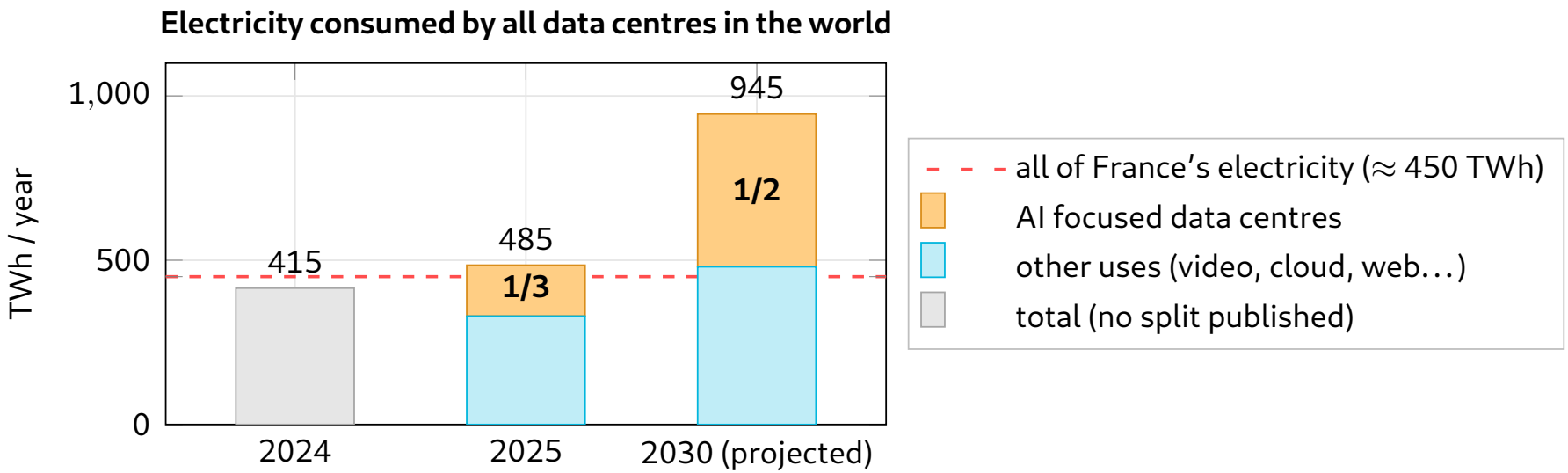
1 Agentic AI: Will AI destroy jobs?

2 Environment or sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate

What AI costs today, and where it is going



1.5%
of world electricity in 2024
(AI alone, 2025: ≈ 0.5%)

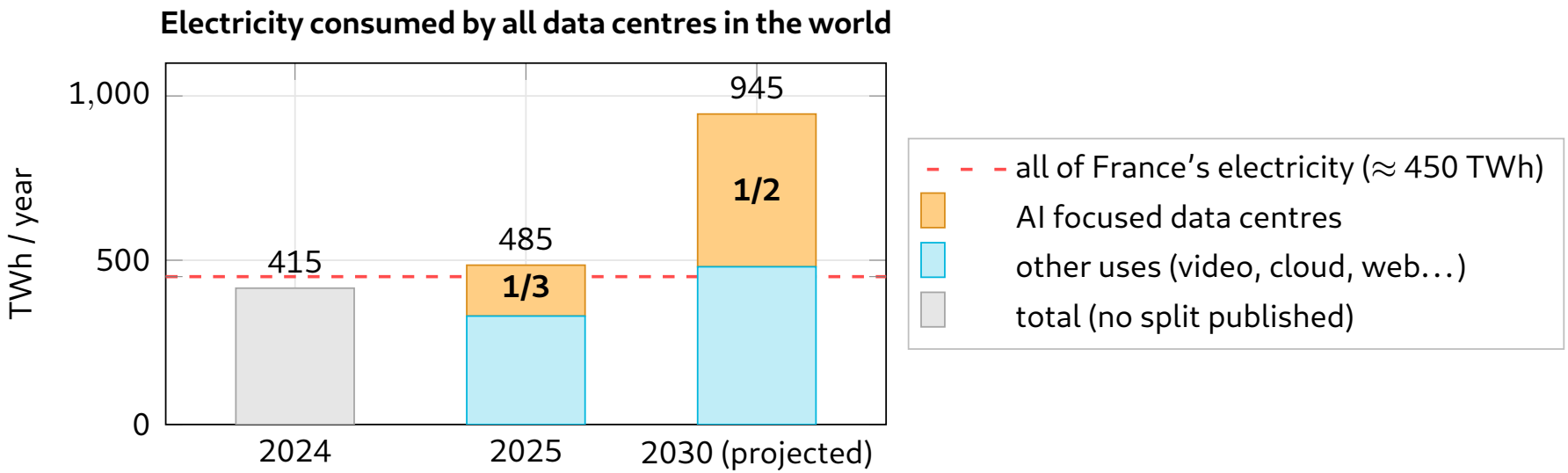
+17%
in 2025 (AI alone: +50%)
all electricity: +3%

×2
by 2030
(AI alone: ×3)

A small share today; but it's one of the fastest growing uses of electricity, and AI drives the growth.

IEA, *Energy and AI*, Apr 2025, `https://www.iea.org/reports/energy-and-ai` IEA, *Key Questions on Energy and AI*, Apr 2026 (AI focused split) Our World in Data, 2026 (AI share of world electricity) RTE for the French figure

What AI costs today, and where it is going



1.5%
of world electricity in 2024
(AI alone, 2025: ≈ 0.5%)

+17%
in 2025 (AI alone: +50%)
all electricity: +3%

×2
by 2030
(AI alone: ×3)

A small share today; but it's one of the fastest growing uses of electricity, and AI drives the growth.

IEA, *Energy and AI*, Apr 2025, `https://www.iea.org/reports/energy-and-ai` IEA, *Key Questions on Energy and AI*, Apr 2026 (AI focused split) Our World in Data, 2026 (AI share of world electricity) RTE for the French figure

One model, measured: Mistral Large 2

Building the model
training, then 18 months of service

20,400 t CO_2e
≈ *the yearly footprint of 2,200 French people*

a large cost, for the model as a whole

+

One more answer
Le Chat, 400 tokens

1.14 g CO_2e

a small cost, at each use

Cost breakdown

Computing (training and inference) account for **85.5%** of the emissions. Mistral does not publish the split between training and inference.
The rest: making the hardware and the buildings.

Training and using one Mistral model costs as much as a small town…
And we don't know for OpenAI, Anthropic, Google, Meta, etc.

Mistral AI, Carbone 4 and ADEME, 22 Jul 2025,
`https://mistral.ai/fr/news/our-contribution-to-a-global-environmental-standard-for-ai/` Equivalences computed with ≈ 9 t CO_2e per French person and per year (SDES) and 2,500 m^3 per Olympic pool

One model, measured: Mistral Large 2

Building the model
training, then 18 months of service

20,400 t CO_2e
≈ *the yearly footprint of 2,200 French people*

a large cost, for the model as a whole

+

One more answer
Le Chat, 400 tokens

1.14 g CO_2e

a small cost, at each use

Cost breakdown

Computing (training and inference) account for **85.5%** of the emissions. Mistral does not publish the split between training and inference.
The rest: making the hardware and the buildings.

Training and using one Mistral model costs as much as a small town…
And we don't know for OpenAI, Anthropic, Google, Meta, etc.

Mistral AI, Carbone 4 and ADEME, 22 Jul 2025,
`https://mistral.ai/fr/news/our-contribution-to-a-global-environmental-standard-for-ai/` Equivalences computed with ≈ 9 t CO_2e per French person and per year (SDES) and 2,500 m^3 per Olympic pool

One model, measured: Mistral Large 2

Building the model
training, then 18 months of service

20,400 t CO_2e
≈ *the yearly footprint of 2,200 French people*

a large cost, for the model as a whole

+

One more answer
Le Chat, 400 tokens

1.14 g CO_2e

a small cost, at each use

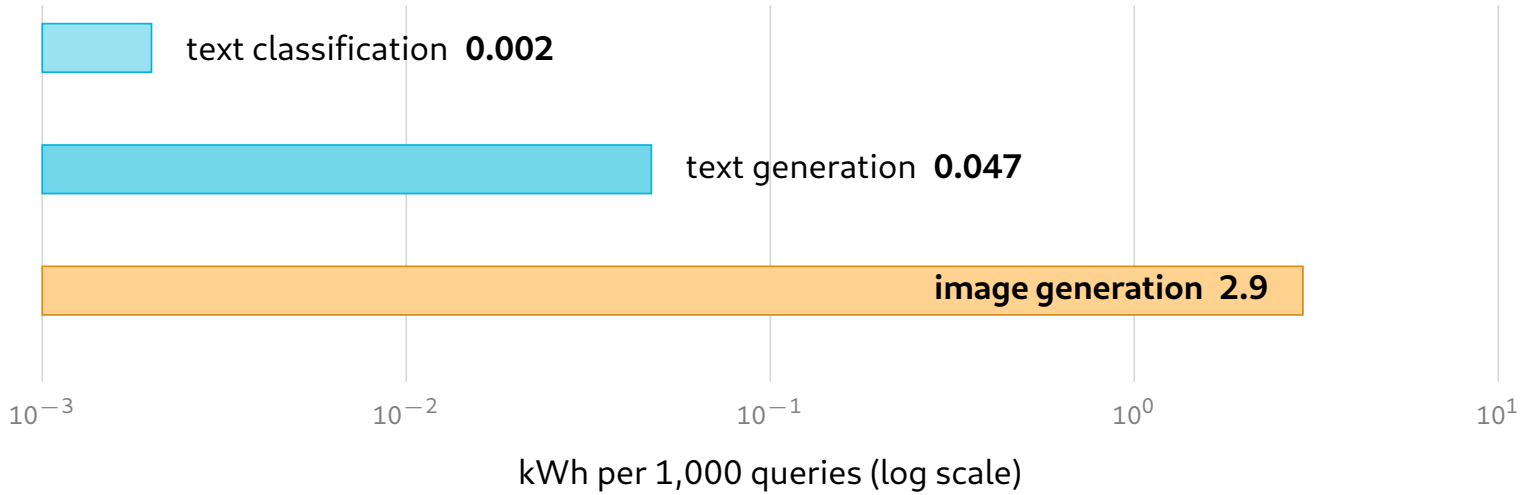
Cost breakdown

Computing (training and inference) account for **85.5%** of the emissions. Mistral does not publish the split between training and inference.
The rest: making the hardware and the buildings.

Training and using one Mistral model costs as much as a small town…
And we don't know for OpenAI, Anthropic, Google, Meta, etc.

Mistral AI, Carbone 4 and ADEME, 22 Jul 2025,
`https://mistral.ai/fr/news/our-contribution-to-a-global-environmental-standard-for-ai/` Equivalences computed with ≈ 9 t CO_2e per French person and per year (SDES) and 2,500 m^3 per Olympic pool

Not all tasks cost the same



× 1,450 between the cheapest and the most expensive task

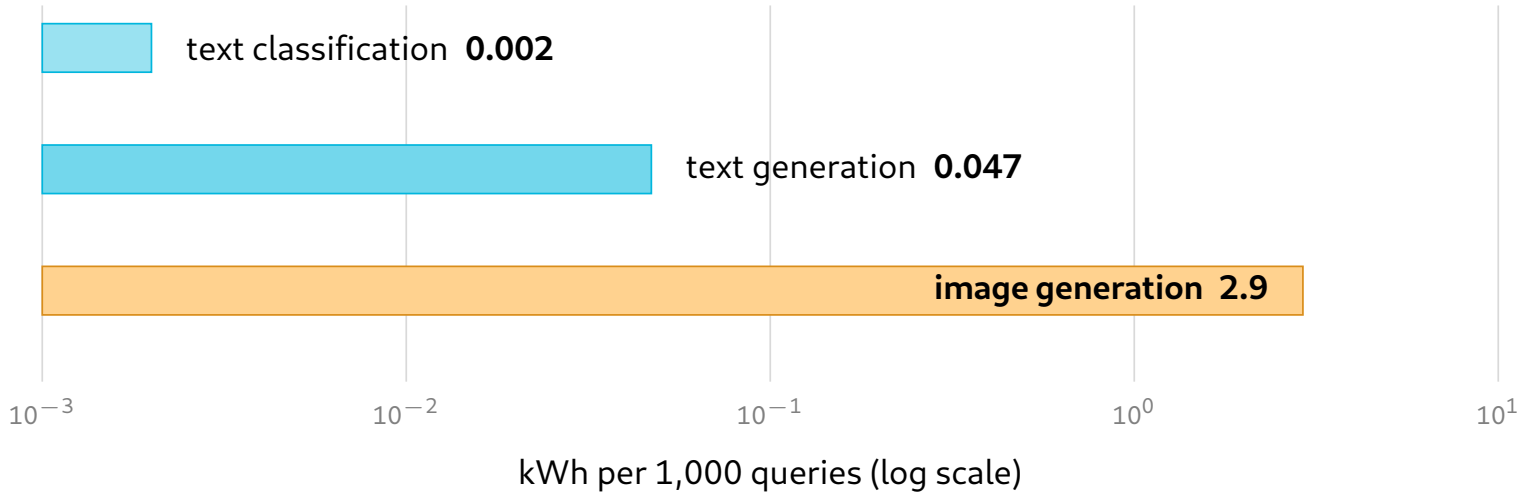
And reasoning models (first part of this course)

A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

"Using AI" is not one activity.

Luccioni, Jernite, Strubell, *Power Hungry Processing*, FAccT 2024, `https://arxiv.org/abs/2311.16863` Oviedo et al., *Joule*, Apr 2026 Google, `https://arxiv.org/abs/2508.15734`

Not all tasks cost the same



× 1,450 between the cheapest and the most expensive task

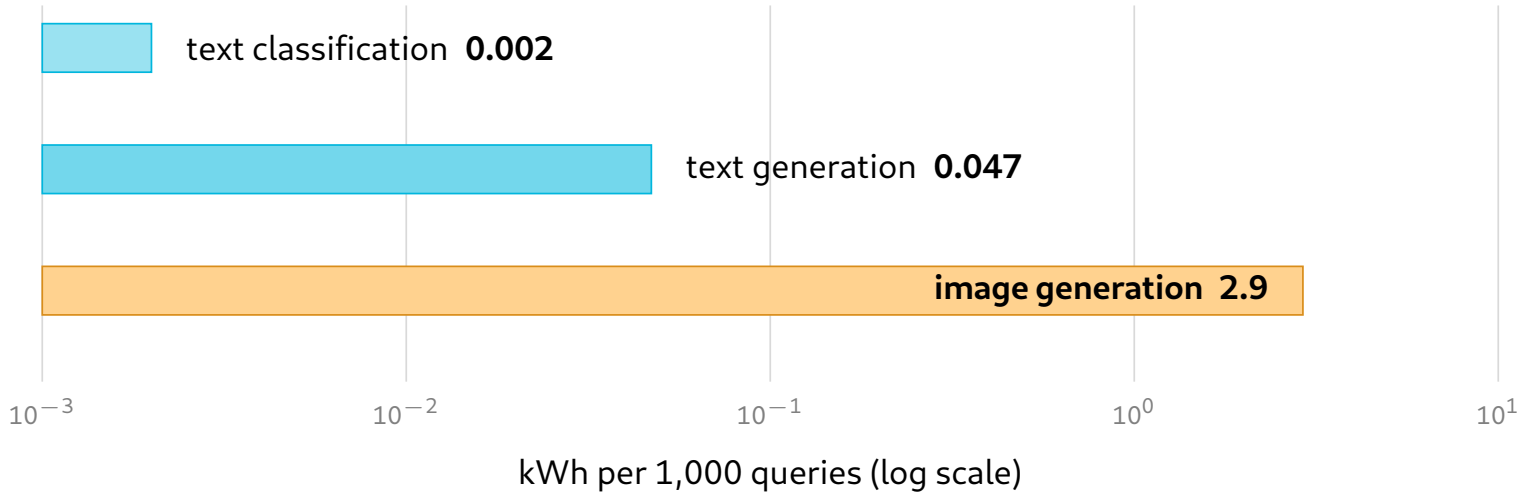
And reasoning models (first part of this course)

A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

"Using AI" is not one activity.

Luccioni, Jernite, Strubell, *Power Hungry Processing*, FAccT 2024, `https://arxiv.org/abs/2311.16863` Oviedo et al., *Joule*, Apr 2026 Google, `https://arxiv.org/abs/2508.15734`

Not all tasks cost the same



× 1,450 between the cheapest and the most expensive task

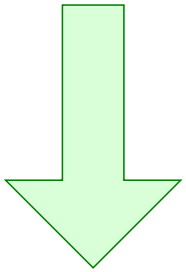
And reasoning models (first part of this course)

A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

"Using AI" is not one activity.

Luccioni, Jernite, Strubell, *Power Hungry Processing*, FAccT 2024, `https://arxiv.org/abs/2311.16863` Oviedo et al., *Joule*, Apr 2026 Google, `https://arxiv.org/abs/2508.15734`

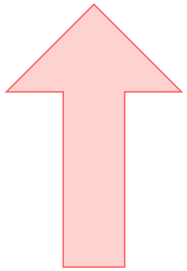
Computing efficiency, and the rebound effect



÷ 33

energy **per prompt** at Google, in 12 months

but



+37%

Google **total** electricity, in 2025

(emissions: +81% since 2019)

Rebound effect (Jevons paradox)

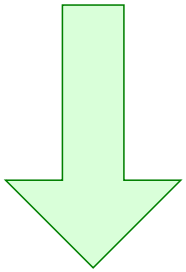
"It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth."
W. S. Jevons, *The Coal Question*, 1865

So: developing and running models is costly

Running: cheaper answers → more answers, in more products, for more people.
Building: Microsoft's largest emissions line is **capital goods** (≈ 43%): the data centres and chips, not the electricity.

Elsworth et al. (Google), *Measuring the environmental impact of delivering AI at Google scale*, Aug 2025, `https://arxiv.org/abs/2508.15734` Google, *2026 Environmental Report*, `https://sustainability.google/reports/google-2026-environmental-report/` Microsoft, *2026 Environmental Data Fact Sheet* (FY2025: capital goods 9.0 of 21.1 Mt CO_2e), `microsoft.com (PDF)` W. S. Jevons, *The Coal Question*, 1865, ch. VII

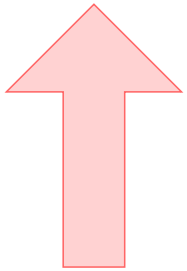
Computing efficiency, and the rebound effect



÷ 33

energy **per prompt** at Google, in 12 months

but



+37%

Google **total** electricity, in 2025

(emissions: +81% since 2019)

Rebound effect (Jevons paradox)

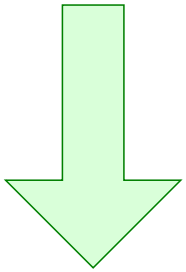
"It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth."
W. S. Jevons, *The Coal Question*, 1865

So: developing and running models is costly

Running: cheaper answers → more answers, in more products, for more people.
Building: Microsoft's largest emissions line is **capital goods** (≈ 43%): the data centres and chips, not the electricity.

Elsworth et al. (Google), *Measuring the environmental impact of delivering AI at Google scale*, Aug 2025, `https://arxiv.org/abs/2508.15734` Google, *2026 Environmental Report*, `https://sustainability.google/reports/google-2026-environmental-report/` Microsoft, *2026 Environmental Data Fact Sheet* (FY2025: capital goods 9.0 of 21.1 Mt CO_2e), `microsoft.com (PDF)` W. S. Jevons, *The Coal Question*, 1865, ch. VII

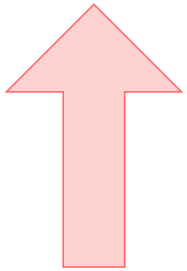
Computing efficiency, and the rebound effect



÷ 33

energy **per prompt** at Google, in 12 months

but



+37%

Google **total** electricity, in 2025

(emissions: +81% since 2019)

Rebound effect (Jevons paradox)

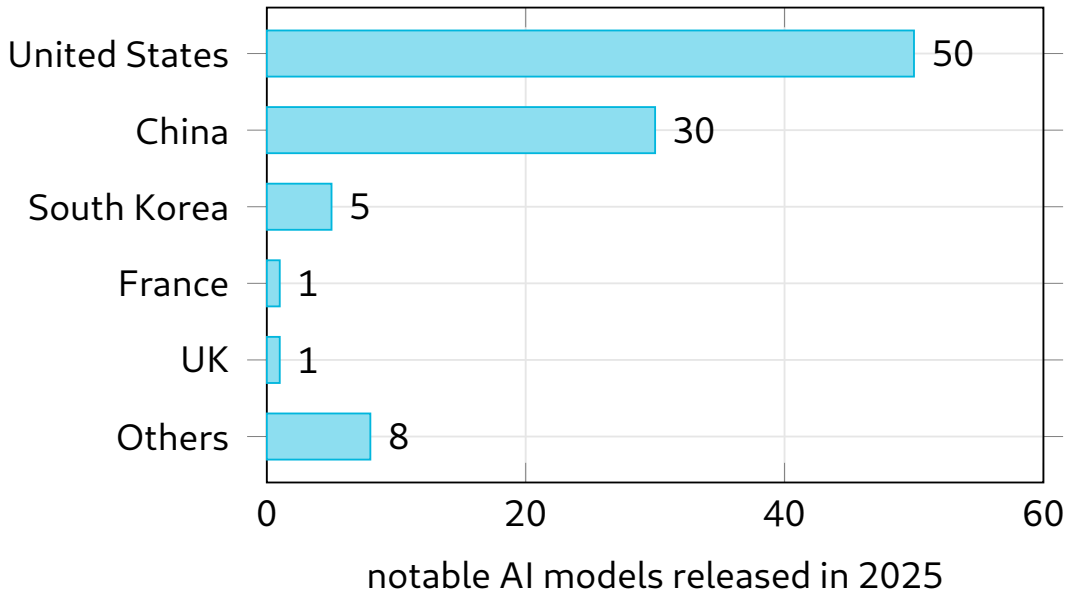
"It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth."
W. S. Jevons, *The Coal Question*, 1865

So: developing and running models is costly

Running: cheaper answers → more answers, in more products, for more people.
Building: Microsoft's largest emissions line is **capital goods** (≈ 43%): the data centres and chips, not the electricity.

Elsworth et al. (Google), *Measuring the environmental impact of delivering AI at Google scale*, Aug 2025, `https://arxiv.org/abs/2508.15734` Google, *2026 Environmental Report*, `https://sustainability.google/reports/google-2026-environmental-report/` Microsoft, *2026 Environmental Data Fact Sheet* (FY2025: capital goods 9.0 of 21.1 Mt CO_2e), `microsoft.com (PDF)` W. S. Jevons, *The Coal Question*, 1865, ch. VII

The frontier: a race between two countries

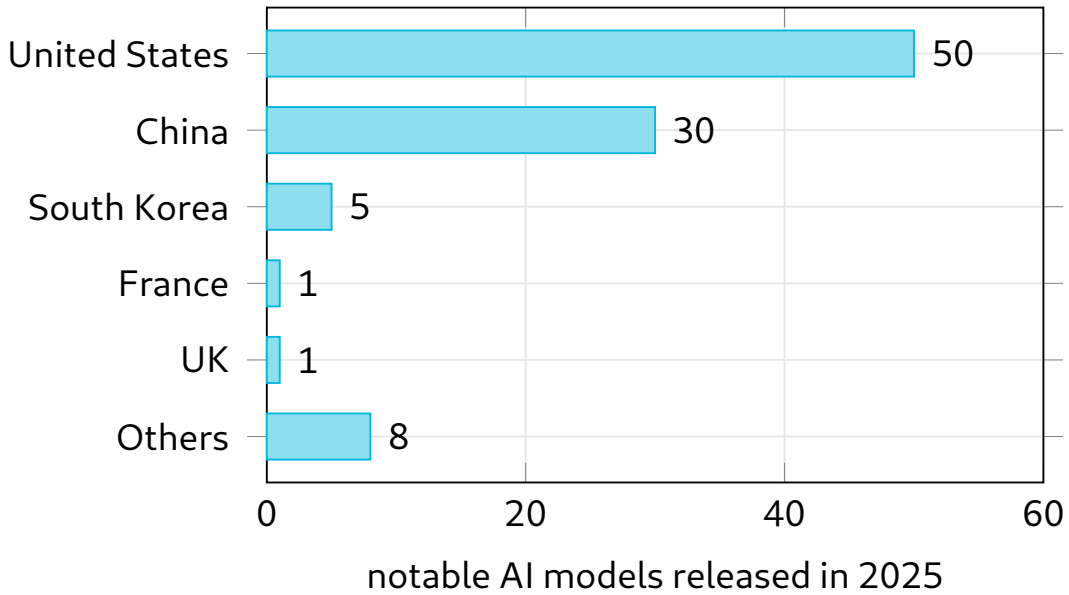


best US model vs
best Chinese model
2.7%
gap in human preference rating
(Arena, March 2026)

80 of the 95 notable models of 2025 are American or Chinese.
France: one.

Stanford HAI, *AI Index 2026*, `https://hai.stanford.edu/ai-index/2026-ai-index-report` (Epoch AI data for notable models, LMArena for the rating)

The frontier: a race between two countries

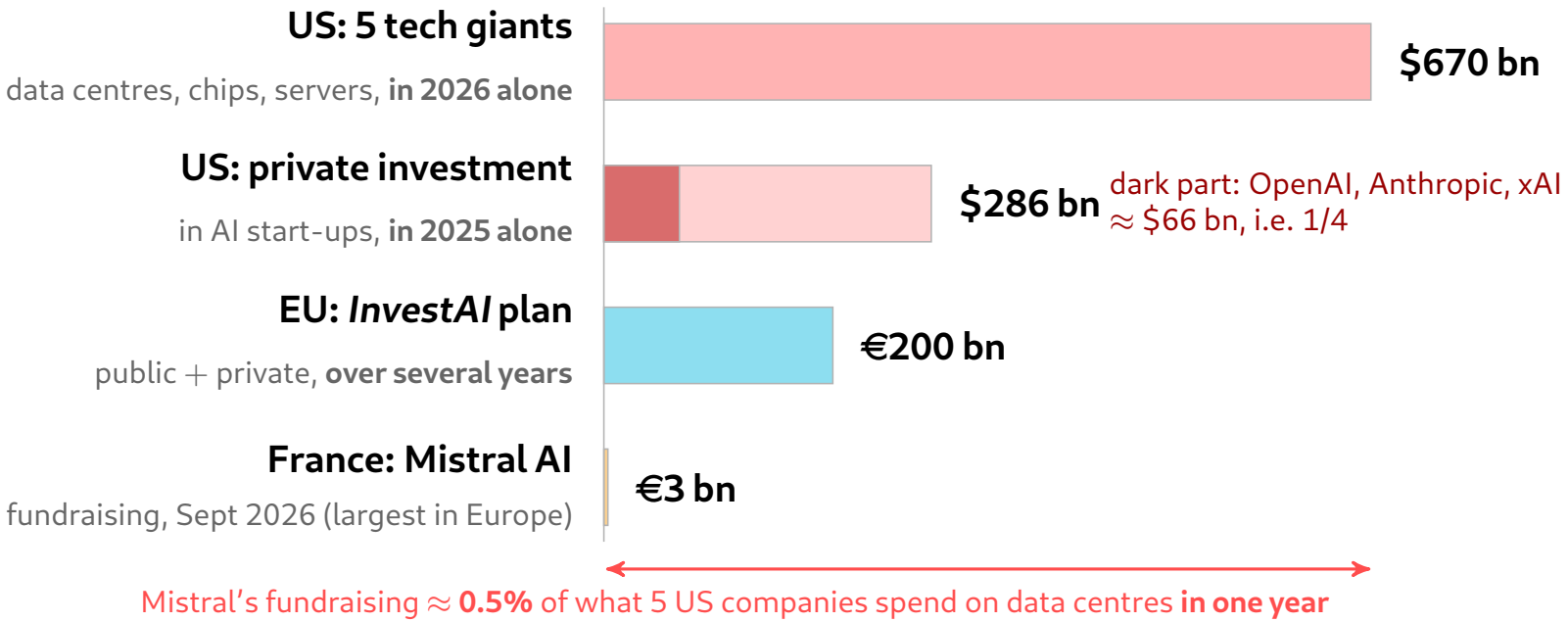


best US model vs
best Chinese model
2.7%
gap in human preference rating
(Arena, March 2026)

80 of the 95 notable models of 2025 are American or Chinese.
France: one.

Stanford HAI, *AI Index 2026*, `https://hai.stanford.edu/ai-index/2026-ai-index-report` (Epoch AI data for notable models, LMArena for the rating)

A ferocious and very costly competition



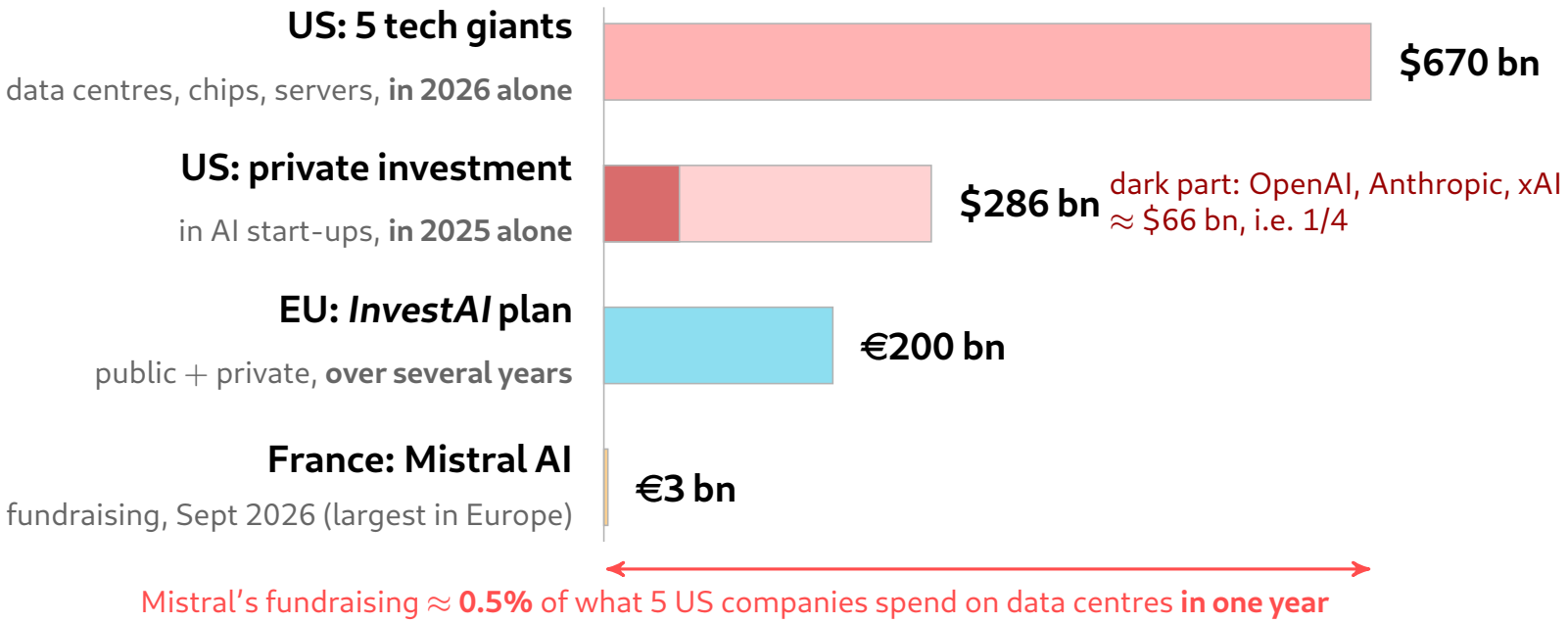
Why it never stops

A frontier model is outdated within months. Staying in the race means **training a bigger model every few months**, on more chips, in larger data centres: the bill grows with each generation.

France (and even Europe) is left behind (for now).

Futurum, 12 Feb 2026, `futurumgroup.com` Stanford HAI, *AI Index 2026*, `hai.stanford.edu` OpenAI, Anthropic: TechCrunch, 19 Jan 2026, `techcrunch.com` xAI: `techfundingnews.com` InvestAI, 11 Feb 2025, `maginative.com` Mistral: TechCrunch, 8 Sept 2026, `techcrunch.com`

A ferocious and very costly competition



Why it never stops

A frontier model is outdated within months. Staying in the race means **training a bigger model every few months**, on more chips, in larger data centres: the bill grows with each generation.

France (and even Europe) is left behind (for now).

Futurum, 12 Feb 2026, `futurumgroup.com` Stanford HAI, *AI Index 2026*, `hai.stanford.edu` OpenAI, Anthropic: TechCrunch, 19 Jan 2026, `techcrunch.com` xAI: `techfundingnews.com` InvestAI, 11 Feb 2025, `maginative.com` Mistral: TechCrunch, 8 Sept 2026, `techcrunch.com`

Sovereignty is not only the AI model!

link	who dominates the world	and Europe?
fabrication	TSMC (Taiwan): **70%** of chip making[1]	**ASML** (NL): sole maker of advanced chip printing machines[2]
chips	NVIDIA (US): >**60%** of AI computing[3]	no European AI chip at scale
cloud	Amazon, Microsoft, Google: **63%**[4]	European providers: **15%** of Europe[5]
data centres	US **45%**, China 25% of world consumption[6]	Europe **15%**[6]; low carbon power in France
models	2025 notable models: US **50**, China **30**[7]	France: **1** (Mistral, open weights)[7]

Five links **in series**

Independence requires all five. And four of them are **physical**: this industry looks like software, and behaves like heavy industry.

Perfect sovereignty would be very expensive.

[1] TrendForce, Q2 2025, `trendforce.com` [2] CNBC, 2022, `cnbc.com` [3] Epoch AI, Jan 2026, `epoch.ai` [4] Synergy Research, Q3 2025, `srgresearch.com` [5] Synergy, via The Register, Jul 2025, `theregister.com` [6] IEA, *Energy and AI*, 2024 data, `iea.org` [7] Stanford HAI, *AI Index 2026*, `hai.stanford.edu`

Sovereignty is not only the AI model!

link	who dominates the world	and Europe?
fabrication	TSMC (Taiwan): **70%** of chip making[1]	**ASML** (NL): sole maker of advanced chip printing machines[2]
chips	NVIDIA (US): >**60%** of AI computing[3]	no European AI chip at scale
cloud	Amazon, Microsoft, Google: **63%**[4]	European providers: **15%** of Europe[5]
data centres	US **45%**, China 25% of world consumption[6]	Europe **15%**[6]; low carbon power in France
models	2025 notable models: US **50**, China **30**[7]	France: **1** (Mistral, open weights)[7]

Five links **in series**

Independence requires all five. And four of them are **physical**: this industry looks like software, and behaves like heavy industry.

Perfect sovereignty would be very expensive.

[1] TrendForce, Q2 2025, `trendforce.com` [2] CNBC, 2022, `cnbc.com` [3] Epoch AI, Jan 2026, `epoch.ai` [4] Synergy Research, Q3 2025, `srgresearch.com` [5] Synergy, via The Register, Jul 2025, `theregister.com` [6] IEA, *Energy and AI*, 2024 data, `iea.org` [7] Stanford HAI, *AI Index 2026*, `hai.stanford.edu`

Already dependent: Europe runs on American Big Tech



Dependance also means loss of control

- **2019 to 2026**: French national health data (Health Data Hub) hosted on Microsoft Azure; move to a French provider decided in April 2026[3],
- **June 2025**, French Senate, under oath: Microsoft France **"cannot guarantee"** that French public data would never be handed to US authorities[4].

The Internet, again? European cloud: **29%** → **15%** of Europe since 2017[1]. **AI could follow.**

Whoever runs the service holds a switch.
And a foreign law, and a foreign president, can reach that switch.

[1] Synergy Research, via The Register, Jul 2025, `theregister.com` [2] US Department of Justice, *CLOUD Act resources*, `justice.gov` [3] Le Monde Informatique, 23 Apr 2026, `lemondeinformatique.fr` [4] French Senate inquiry on public procurement, Jun 2025, via SDxCentral, `sdxcentral.com`

Already dependent: Europe runs on American Big Tech



Dependance also means loss of control

- **2019 to 2026**: French national health data (Health Data Hub) hosted on Microsoft Azure; move to a French provider decided in April 2026[3],
- **June 2025**, French Senate, under oath: Microsoft France **"cannot guarantee"** that French public data would never be handed to US authorities[4].

The Internet, again? European cloud: **29%** → **15%** of Europe since 2017[1]. **AI could follow.**

Whoever runs the service holds a switch.
And a foreign law, and a foreign president, can reach that switch.

[1] Synergy Research, via The Register, Jul 2025, `theregister.com` [2] US Department of Justice, *CLOUD Act resources*, `justice.gov` [3] Le Monde Informatique, 23 Apr 2026, `lemondeinformatique.fr` [4] French Senate inquiry on public procurement, Jun 2025, via SDxCentral, `sdxcentral.com`

Already dependent: Europe runs on American Big Tech



Dependance also means loss of control

- **2019 to 2026**: French national health data (Health Data Hub) hosted on Microsoft Azure; move to a French provider decided in April 2026[3],
- **June 2025**, French Senate, under oath: Microsoft France **"cannot guarantee"** that French public data would never be handed to US authorities[4].

The Internet, again? European cloud: **29%** → **15%** of Europe since 2017[1]. **AI could follow.**

Whoever runs the service holds a switch.
And a foreign law, and a foreign president, can reach that switch.

[1] Synergy Research, via The Register, Jul 2025, `theregister.com` [2] US Department of Justice, *CLOUD Act resources*, `justice.gov` [3] Le Monde Informatique, 23 Apr 2026, `lemondeinformatique.fr` [4] French Senate inquiry on public procurement, Jun 2025, via SDxCentral, `sdxcentral.com`

Already dependent: Europe runs on American Big Tech



Dependance also means loss of control

- **2019 to 2026**: French national health data (Health Data Hub) hosted on Microsoft Azure; move to a French provider decided in April 2026[3],
- **June 2025**, French Senate, under oath: Microsoft France **"cannot guarantee"** that French public data would never be handed to US authorities[4].

The Internet, again? European cloud: **29%** → **15%** of Europe since 2017[1]. **AI could follow.**

Whoever runs the service holds a switch.
And a foreign law, and a foreign president, can reach that switch.

[1] Synergy Research, via The Register, Jul 2025, `theregister.com` [2] US Department of Justice, *CLOUD Act resources*, `justice.gov` [3] Le Monde Informatique, 23 Apr 2026, `lemondeinformatique.fr` [4] French Senate inquiry on public procurement, Jun 2025, via SDxCentral, `sdxcentral.com`

Place yourself: should France build its own frontier model?

pick one side

YES *or* **NO**

Strongest for "yes"

- Already dependent: **70%** of European cloud is American (CLOUD Act),
- The Internet lesson: European cloud **29%** $\rightarrow$ **15%**,
- Notable models in 2025: US **50**, China **30**, France **1**,
- Real assets: ASML, Mistral, low carbon power.

Strongest for "no"

- One model $\approx$ yearly CO_2 of **2,200** French people,
- Rebound effect: per prompt $\div$**33**, total **+37%**,
- Mistral raises **0.5%** of one year of US spending,
- A model is 1 link out of 5: still dependent.

Sources: previous slides

Outline

1 Agentic AI: Will AI destroy jobs?

2 Environment or sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate

Generating art from one sentence is now easy.

Who needed it? And who pays for it?



	AI as a tool: the human chooses		AI as a generator: the model chooses
images	restoration, upscaling, removing a background	→	a picture from one sentence (Midjourney)
text	spelling, translation, rephrasing	→	a whole story from a prompt
voice	noise removal, restoring old recordings	→	a cloned voice reading any text
video	rotoscoping, stabilisation, colour grading	→	a clip from a prompt (Sora, Veo)
music	stem separation, mastering, timbre transfer	→	a full song from one sentence (Suno, Udio)

Generating art from one sentence is now easy.

Who needed it? And who pays for it?



	AI as a tool: the human chooses		AI as a generator: the model chooses
images	restoration, upscaling, removing a background	→	a picture from one sentence (Midjourney)
text	spelling, translation, rephrasing	→	a whole story from a prompt
voice	noise removal, restoring old recordings	→	a cloned voice reading any text
video	rotoscoping, stabilisation, colour grading	→	a clip from a prompt (Sora, Veo)
music	stem separation, mastering, timbre transfer	→	a full song from one sentence (Suno, Udio)

What AI music enables

the human makes the choices — **the model makes the choices**

Restore
The Beatles, *Now and Then* (2023)
Lennon's voice **separated** from an old cassette (separation, not generation). Grammy 2025

Tools for musicians
Stems, mastering, timbres
separate the instruments of a track, master a demo, turn your voice into a violin (RAVE, IRCAM, 2021)

Give a voice back
Randy Travis (2024)
voiceless after a stroke, releases a new song: his voice recreated by AI, **with his consent**

New creators
Xania Monet (2025)
a human writes the lyrics, Suno makes the music: $3M record deal. Suno: **2 million** paying users (2026)

Some Music was never meant to be art

Jingles, video backgrounds, game loops: **cheap music** for small creators who could never pay a composer.

Some Music was never meant to be serious

A song for a friend's birthday, a joke with your band: like karaoke, the point is **the fun of making**, not the art.

Different models and usage means different ethical considerations.

Randy Travis, May 2024 Xania Monet, Sep 2025 Suno, Feb 2026 (company figure) RAVE: Caillon & Esling, `https://arxiv.org/abs/2111.05011`

What AI music enables

the human makes the choices → **the model makes the choices**

Restore
The Beatles, *Now and Then* (2023)
Lennon's voice **separated** from an old cassette (separation, not generation). Grammy 2025

Tools for musicians
Stems, mastering, timbres
separate the instruments of a track, master a demo, turn your voice into a violin (RAVE, IRCAM, 2021)

Give a voice back
Randy Travis (2024)
voiceless after a stroke, releases a new song: his voice recreated by AI, **with his consent**

New creators
Xania Monet (2025)
a human writes the lyrics, Suno makes the music: $3M record deal. Suno: **2 million** paying users (2026)

Some Music was never meant to be art

Jingles, video backgrounds, game loops: **cheap music** for small creators who could never pay a composer.

Some Music was never meant to be serious

A song for a friend's birthday, a joke with your band: like karaoke, the point is **the fun of making**, not the art.

Different models and usage means different ethical considerations.

Randy Travis, May 2024 Xania Monet, Sep 2025 Suno, Feb 2026 (company figure) RAVE: Caillon & Esling, `https://arxiv.org/abs/2111.05011`

What AI music enables

the human makes the choices — **the model makes the choices**

Restore
The Beatles, *Now and Then* (2023)
Lennon's voice **separated** from an old cassette (separation, not generation). Grammy 2025

Tools for musicians
Stems, mastering, timbres
separate the instruments of a track, master a demo, turn your voice into a violin (RAVE, IRCAM, 2021)

Give a voice back
Randy Travis (2024)
voiceless after a stroke, releases a new song: his voice recreated by AI, **with his consent**

New creators
Xania Monet (2025)
a human writes the lyrics, Suno makes the music: $3M record deal. Suno: **2 million** paying users (2026)

Some Music was never meant to be art

Jingles, video backgrounds, game loops: **cheap music** for small creators who could never pay a composer.

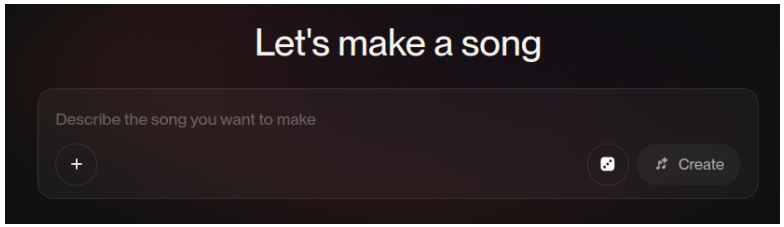
Some Music was never meant to be serious

A song for a friend's birthday, a joke with your band: like karaoke, the point is **the fun of making**, not the art.

Different models and usage means different ethical considerations.

Randy Travis, May 2024 Xania Monet, Sep 2025 Suno, Feb 2026 (company figure) RAVE: Caillon & Esling, `https://arxiv.org/abs/2111.05011`

Data and model: the example of Suno



"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

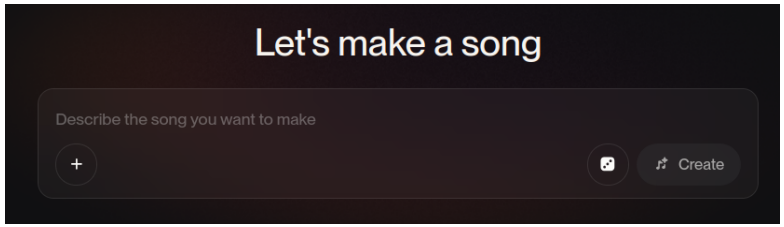
And in Europe?

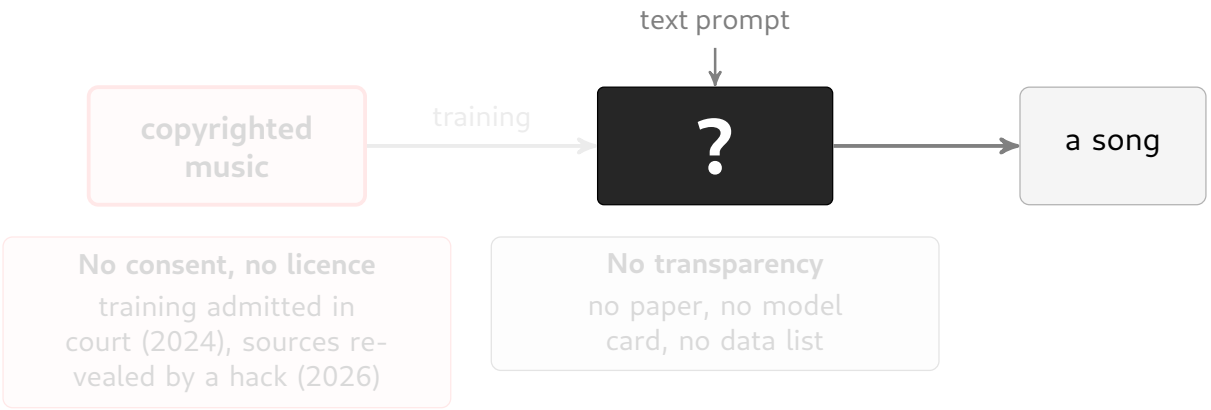
Training allowed **unless** rights holders opt out (2019). AI Act: a **summary of training data** mandatory since Aug 2025.

We cannot audit the model: the debate is built on lawsuits and leaks.

Suno's filing (UMG v. Suno), Aug 2024 404 Media, 15 Jul 2026, `https://www.404media.co/hack-reveals-suno-ai-music-generator-scraped-youtube-deezer-and-genius/` GEMA v. Suno, LG München I, 31 Jul 2026 (under appeal) Directive (EU) 2019/790, Art. 4 AI Act, Art. 53

Data and model: the example of Suno





"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

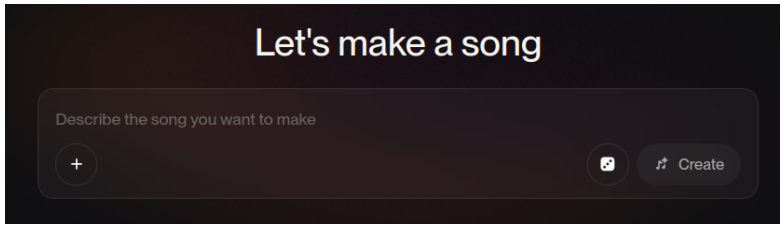
And in Europe?

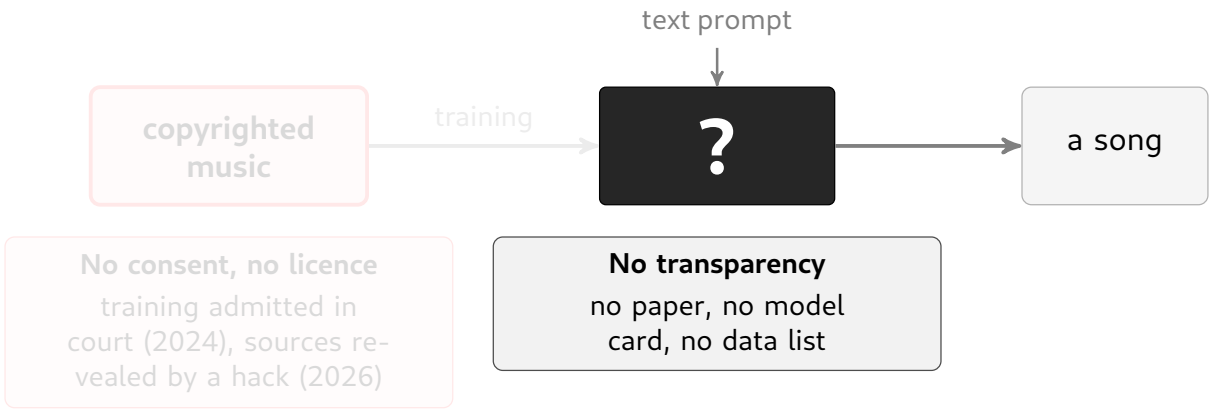
Training allowed **unless** rights holders opt out (2019). AI Act: a **summary of training data** mandatory since Aug 2025.

We cannot audit the model: the debate is built on lawsuits and leaks.

Suno's filing (UMG v. Suno), Aug 2024 404 Media, 15 Jul 2026, `https://www.404media.co/hack-reveals-suno-ai-music-generator-scraped-youtube-deezer-and-genius/` GEMA v. Suno, LG München I, 31 Jul 2026 (under appeal) Directive (EU) 2019/790, Art. 4 AI Act, Art. 53

Data and model: the example of Suno





"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

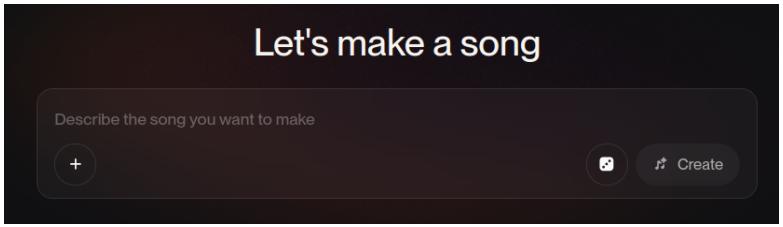
And in Europe?

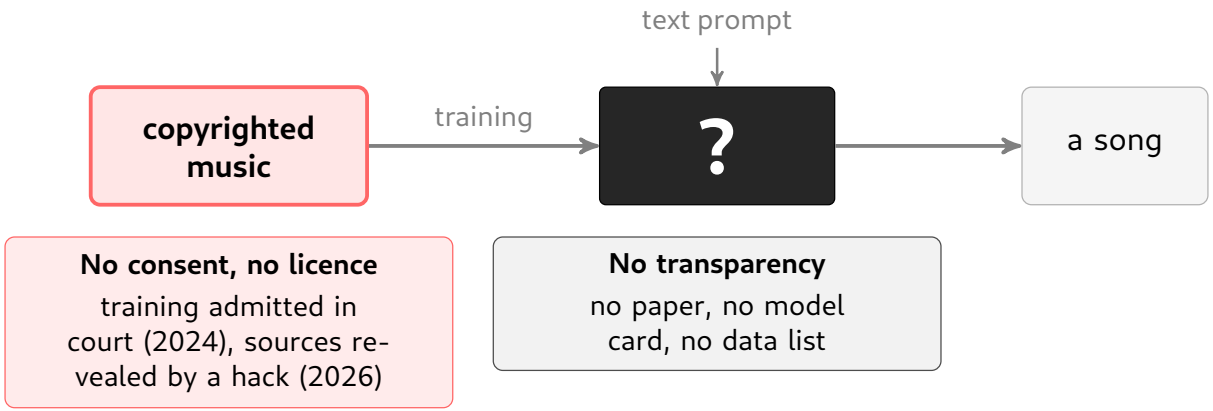
Training allowed **unless** rights holders opt out (2019). AI Act: a **summary of training data** mandatory since Aug 2025.

We cannot audit the model: the debate is built on lawsuits and leaks.

Suno's filing (UMG v. Suno), Aug 2024 404 Media, 15 Jul 2026, `https://www.404media.co/hack-reveals-suno-ai-music-generator-scraped-youtube-deezer-and-genius/` GEMA v. Suno, LG München I, 31 Jul 2026 (under appeal) Directive (EU) 2019/790, Art. 4 AI Act, Art. 53

Data and model: the example of Suno





"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

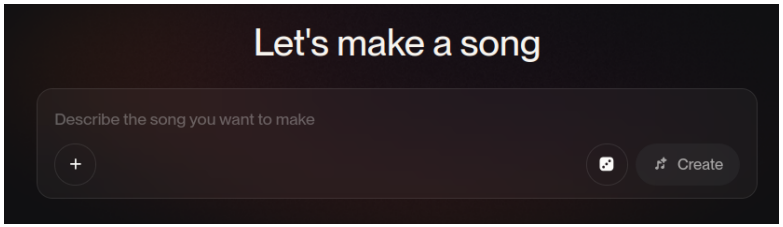
And in Europe?

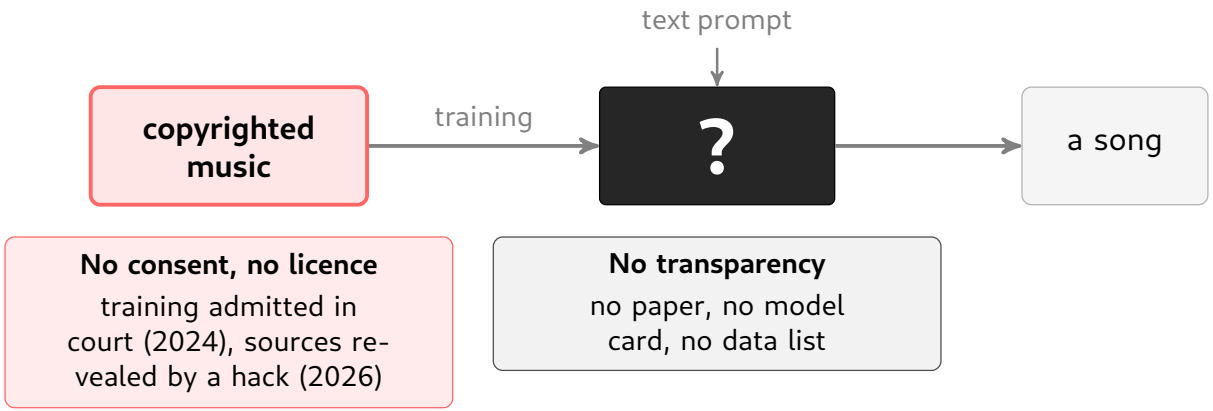
Training allowed **unless** rights holders opt out (2019). AI Act: a **summary of training data** mandatory since Aug 2025.

We cannot audit the model: the debate is built on lawsuits and leaks.

Suno's filing (UMG v. Suno), Aug 2024 404 Media, 15 Jul 2026, `https://www.404media.co/hack-reveals-suno-ai-music-generator-scraped-youtube-deezer-and-genius/` GEMA v. Suno, LG München I, 31 Jul 2026 (under appeal) Directive (EU) 2019/790, Art. 4 AI Act, Art. 53

Data and model: the example of Suno





"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

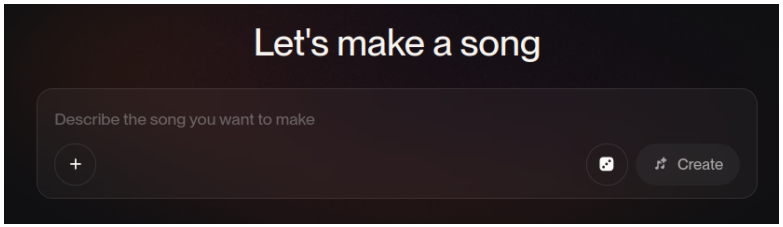
And in Europe?

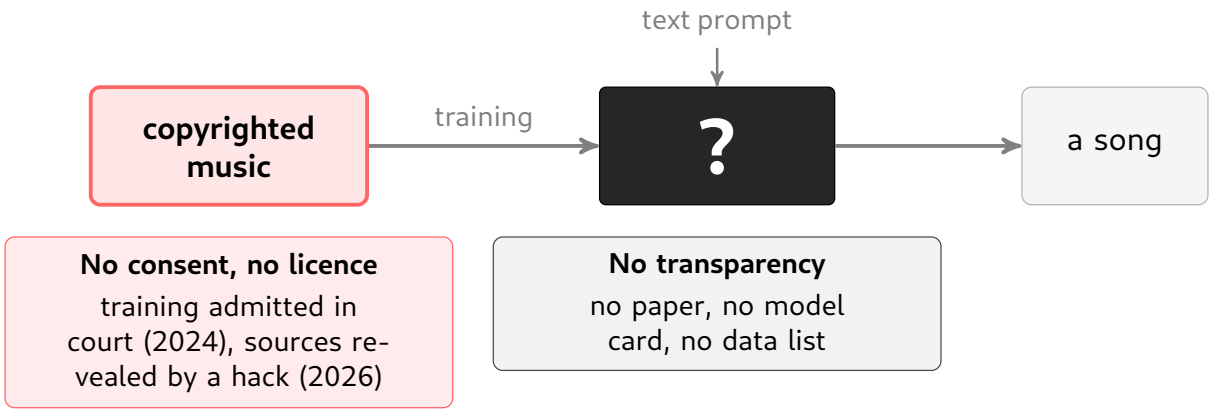
Training allowed **unless** rights holders opt out (2019). AI Act: a **summary of training data** mandatory since Aug 2025.

We cannot audit the model: the debate is built on lawsuits and leaks.

Suno's filing (UMG v. Suno), Aug 2024 404 Media, 15 Jul 2026, `https://www.404media.co/hack-reveals-suno-ai-music-generator-scraped-youtube-deezer-and-genius/` GEMA v. Suno, LG München I, 31 Jul 2026 (under appeal) Directive (EU) 2019/790, Art. 4 AI Act, Art. 53

Data and model: the example of Suno





"But humans learn from music too!"

True, but a model differs in **scale** (all recorded music, in weeks), it can **memorise** (Munich, 2026: GEMA wins against Suno), and it **substitutes**: it sells songs competing with its sources.

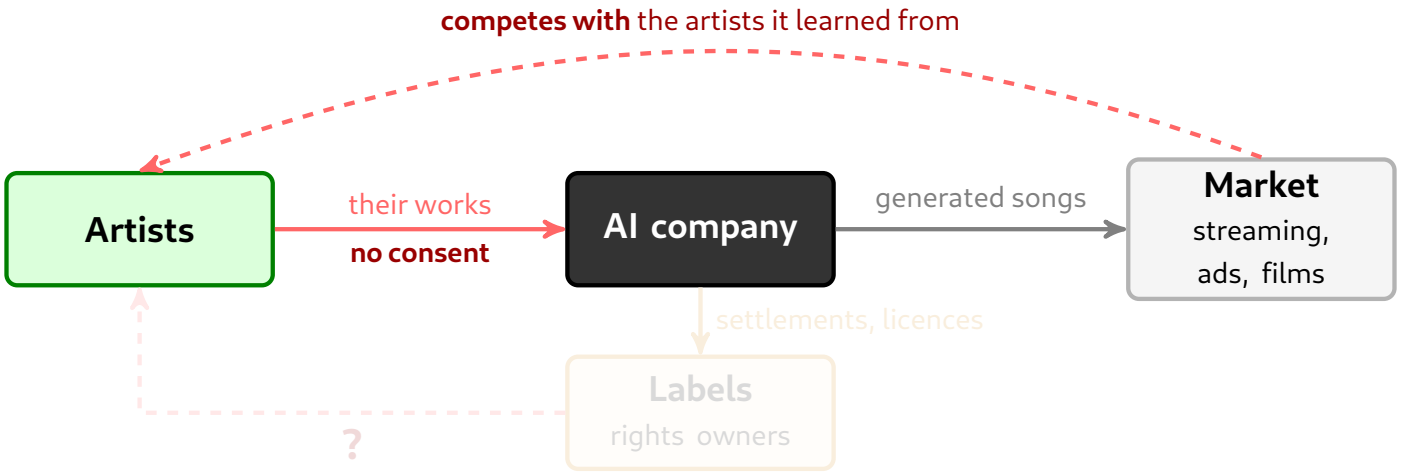
And in Europe?

Training allowed **unless** rights holders opt out (2019). AI Act: a **summary of training data** mandatory since Aug 2025.

We cannot audit the model: the debate is built on lawsuits and leaks.

Suno's filing (UMG v. Suno), Aug 2024 404 Media, 15 Jul 2026, `https://www.404media.co/hack-reveals-suno-ai-music-generator-scraped-youtube-deezer-and-genius/` GEMA v. Suno, LG München I, 31 Jul 2026 (under appeal) Directive (EU) 2019/790, Art. 4 AI Act, Art. 53

Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

Holly Herndon (2021): her voice as an open model.
Grimes (2023): use my voice, **50%** of royalties.
Warner + Suno (2025): artists **opt in**.

Said no

200+ artists, open letter (2024): Billie Eilish, Pearl Jam.
1,000+ musicians, silent album *Is This What We Want?* (2025).

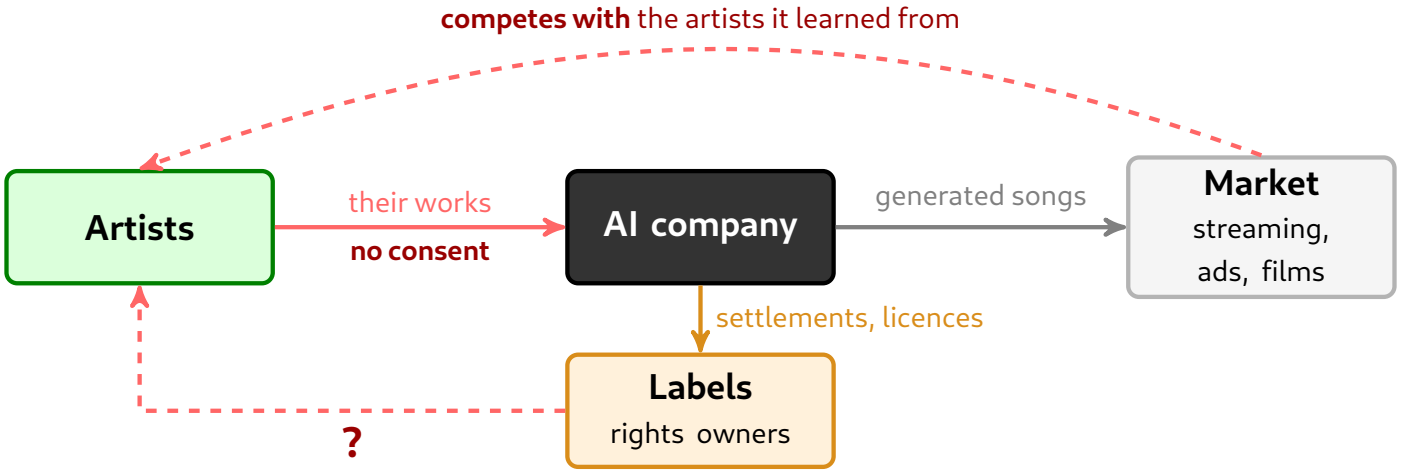
Never asked

Fake Drake (2023): a cloned voice, millions of plays, then removed.
Session musicians sue Universal and Warner (2026): no share of the deals.

An inequality between artists who can negotiate, and those who cannot.

Settlements: Udio and Universal (Oct 2025), Suno and Warner (Nov 2025) Artist Rights Alliance letter, Apr 2024 `https://www.isthiswhatwewant.com/` *Heart on My Sleeve*, Apr 2023 Tennessee ELVIS Act, Mar 2024 AFM v. UMG and Warner, S.D.N.Y., Jun 2026

Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

Holly Herndon (2021): her voice as an open model.
Grimes (2023): use my voice, **50%** of royalties.
Warner + Suno (2025): artists **opt in**.

Said no

200+ artists, open letter (2024): Billie Eilish, Pearl Jam.
1,000+ musicians, silent album *Is This What We Want?* (2025).

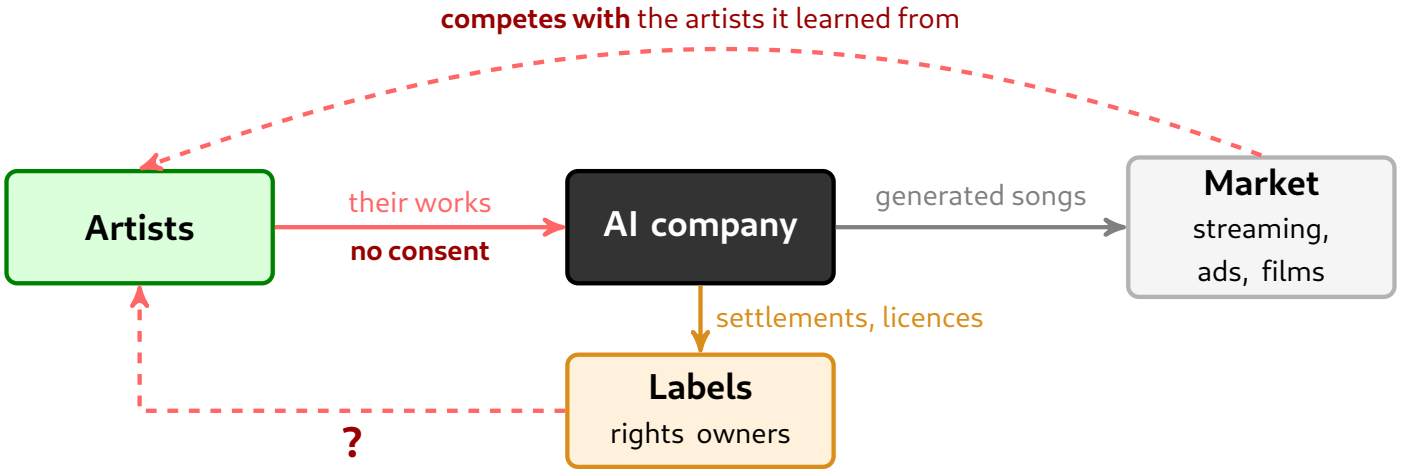
Never asked

Fake Drake (2023): a cloned voice, millions of plays, then removed.
Session musicians sue Universal and Warner (2026): no share of the deals.

An inequality between artists who can negotiate, and those who cannot.

Settlements: Udio and Universal (Oct 2025), Suno and Warner (Nov 2025) Artist Rights Alliance letter, Apr 2024 `https://www.isthiswhatwewant.com/` *Heart on My Sleeve*, Apr 2023 Tennessee ELVIS Act, Mar 2024 AFM v. UMG and Warner, S.D.N.Y., Jun 2026

Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

Holly Herndon (2021): her voice as an open model.
Grimes (2023): use my voice, **50%** of royalties.
Warner + Suno (2025): artists **opt in**.

Said no

200+ artists, open letter (2024): Billie Eilish, Pearl Jam.
1,000+ musicians, silent album *Is This What We Want?* (2025).

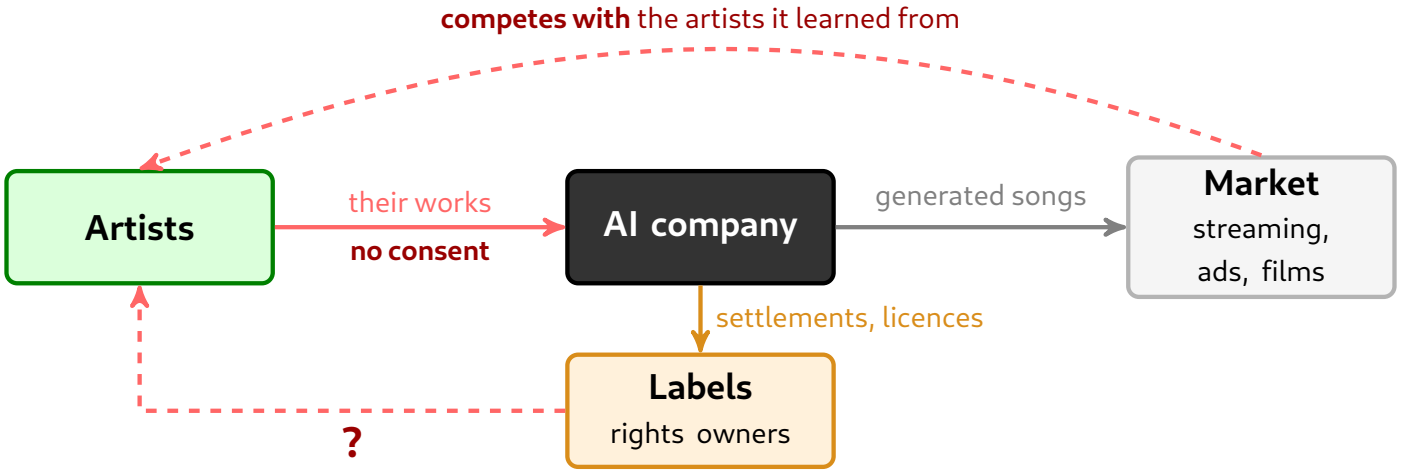
Never asked

Fake Drake (2023): a cloned voice, millions of plays, then removed.
Session musicians sue Universal and Warner (2026): no share of the deals.

An inequality between artists who can negotiate, and those who cannot.

Settlements: Udio and Universal (Oct 2025), Suno and Warner (Nov 2025) Artist Rights Alliance letter, Apr 2024 `https://www.isthiswhatwewant.com/` *Heart on My Sleeve*, Apr 2023 Tennessee ELVIS Act, Mar 2024 AFM v. UMG and Warner, S.D.N.Y., Jun 2026

Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

Holly Herndon (2021): her voice as an open model.
Grimes (2023): use my voice, **50%** of royalties.
Warner + Suno (2025): artists **opt in**.

Said no

200+ artists, open letter (2024): Billie Eilish, Pearl Jam.
1,000+ musicians, silent album *Is This What We Want?* (2025).

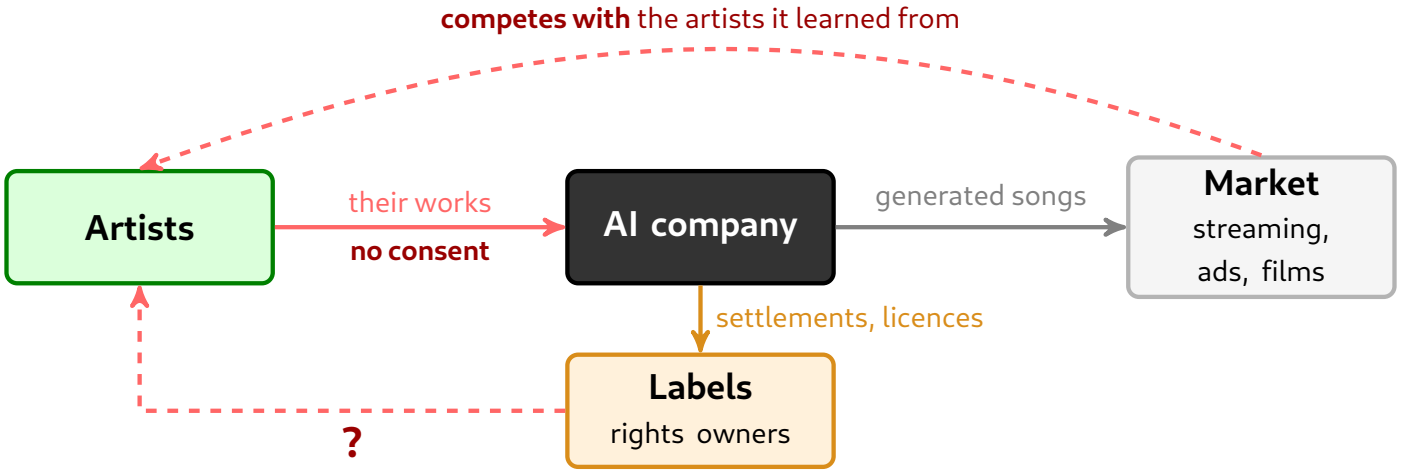
Never asked

Fake Drake (2023): a cloned voice, millions of plays, then removed.
Session musicians sue Universal and Warner (2026): no share of the deals.

An inequality between artists who can negotiate, and those who cannot.

Settlements: Udio and Universal (Oct 2025), Suno and Warner (Nov 2025) Artist Rights Alliance letter, Apr 2024 `https://www.isthiswhatwewant.com/` *Heart on My Sleeve*, Apr 2023 Tennessee ELVIS Act, Mar 2024 AFM v. UMG and Warner, S.D.N.Y., Jun 2026

Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

Holly Herndon (2021): her voice as an open model.
Grimes (2023): use my voice, **50%** of royalties.
Warner + Suno (2025): artists **opt in**.

Said no

200+ artists, open letter (2024): Billie Eilish, Pearl Jam.
1,000+ musicians, silent album *Is This What We Want?* (2025).

Never asked

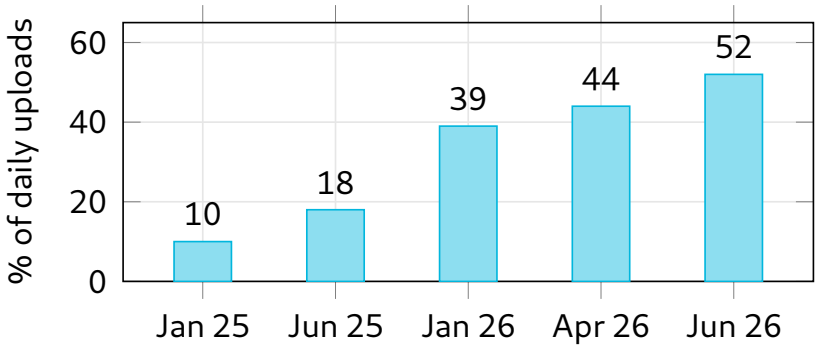
Fake Drake (2023): a cloned voice, millions of plays, then removed.
Session musicians sue Universal and Warner (2026): no share of the deals.

An inequality between artists who can negotiate, and those who cannot.

Settlements: Udio and Universal (Oct 2025), Suno and Warner (Nov 2025) Artist Rights Alliance letter, Apr 2024 `https://www.isthiswhatwewant.com/` *Heart on My Sleeve*, Apr 2023 Tennessee ELVIS Act, Mar 2024 AFM v. UMG and Warner, S.D.N.Y., Jun 2026

The platforms are already flooded with AI music!

Flooded: fully AI generated uploads, Deezer



≈ **90,000 tracks a day** (Oct 2026)

company data, one platform: not generalizable

Barely listened: only **1 to 3%** of all streams

of which up to 85% bots (2025) | humans

bots stream the tracks to collect royalties: **fraud**

cause, correlation, or neither?

Deezer filters (own detector): AI tracks **excluded** from recommendations and playlists, bot streams **not paid**

Some AI bands make it

The Velvet Sundown (2025): ≈ **1M** monthly listeners on Spotify, then revealed as AI

97% fail a blind test (human or AI?)

80% want AI music labelled

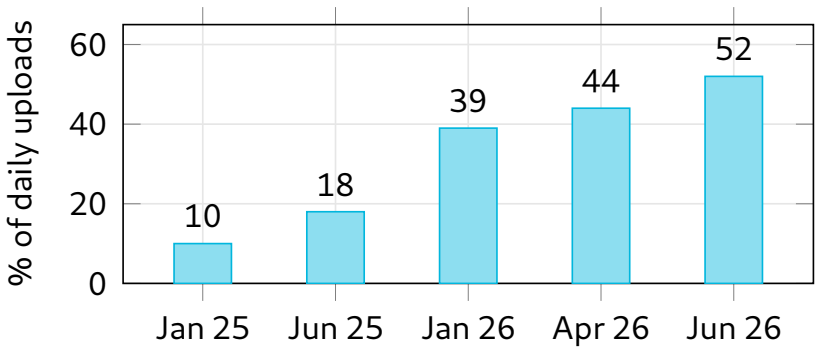
Spotify (2026)

introduces an "AI Persona" label (it was ambiguous about AI generated music before)

Deezer Newsroom, Jul 2026 Music Business Worldwide, 13 Oct 2026 US v. Michael Smith, S.D.N.Y., Sep 2024 *The Velvet Sundown*, Jul 2025 Deezer and Ipsos, Nov 2025 Spotify announcement, Aug 2026 (via Hypebot)

The platforms are already flooded with AI music!

Flooded: fully AI generated uploads, Deezer



≈ **90,000 tracks a day** (Oct 2026)

company data, one platform: not generalizable

Barely listened: only **1 to 3%** of all streams



bots stream the tracks to collect royalties: **fraud**

cause, correlation, or neither?

Deezer filters (own detector): AI tracks **excluded** from recommendations and playlists, bot streams **not paid**

Some AI bands make it

The Velvet Sundown (2025): ≈ **1M** monthly listeners on Spotify, then revealed as AI

97% fail a blind test (human or AI?)

80% want AI music labelled

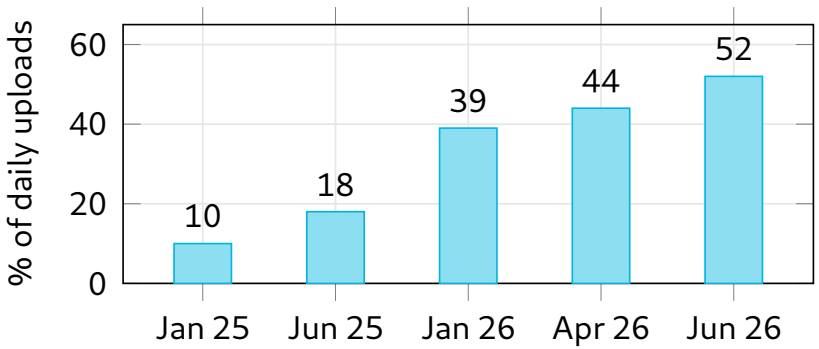
Spotify (2026)

introduces an "AI Persona" label (it was ambiguous about AI generated music before)

Deezer Newsroom, Jul 2026 Music Business Worldwide, 13 Oct 2026 US v. Michael Smith, S.D.N.Y., Sep 2024 *The Velvet Sundown*, Jul 2025 Deezer and Ipsos, Nov 2025 Spotify announcement, Aug 2026 (via Hypebot)

The platforms are already flooded with AI music!

Flooded: fully AI generated uploads, Deezer



≈ **90,000 tracks a day** (Oct 2026)

company data, one platform: not generalizable

Barely listened: only **1 to 3%** of all streams



bots stream the tracks to collect royalties: **fraud**

cause, correlation, or neither?

Deezer filters (own detector): AI tracks **excluded** from recommendations and playlists, bot streams **not paid**

Some AI bands make it

The Velvet Sundown (2025): ≈ **1M** monthly listeners on Spotify, then revealed as AI

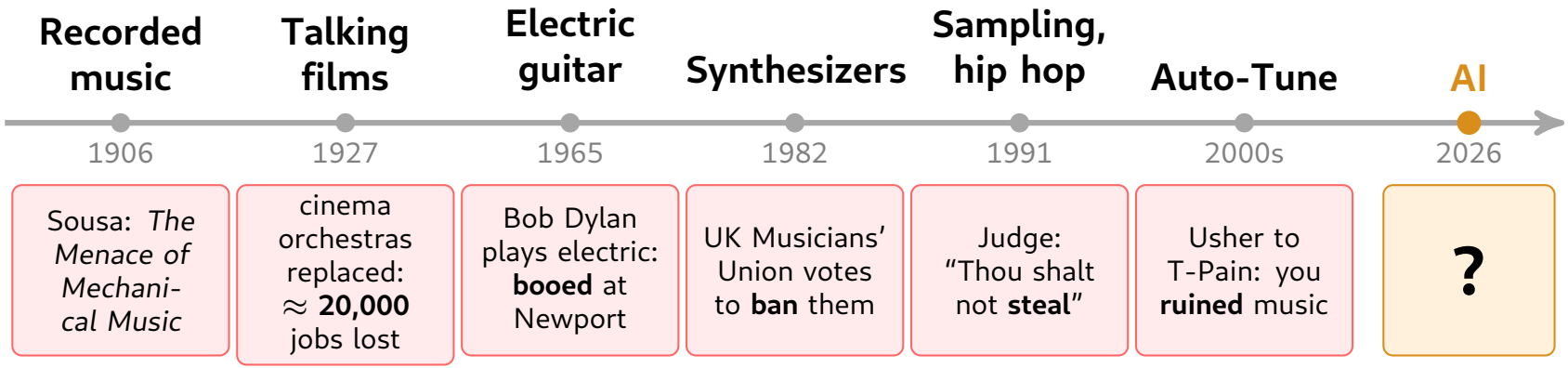
97% fail a blind test (human or AI?)

80% want AI music labelled

Spotify (2026) introduces an "AI Persona" label (it was ambiguous about AI generated music before)

Deezer Newsroom, Jul 2026 Music Business Worldwide, 13 Oct 2026 US v. Michael Smith, S.D.N.Y., Sep 2024 *The Velvet Sundown*, Jul 2025 Deezer and Ipsos, Nov 2025 Spotify announcement, Aug 2026 (via Hypebot)

Is AI the problem, or the next instrument?



first seen as a threat to music ⟶ all mainstream today
but "mainstream" does not mean "nobody lost": some fears were right

The sampling precedent

After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

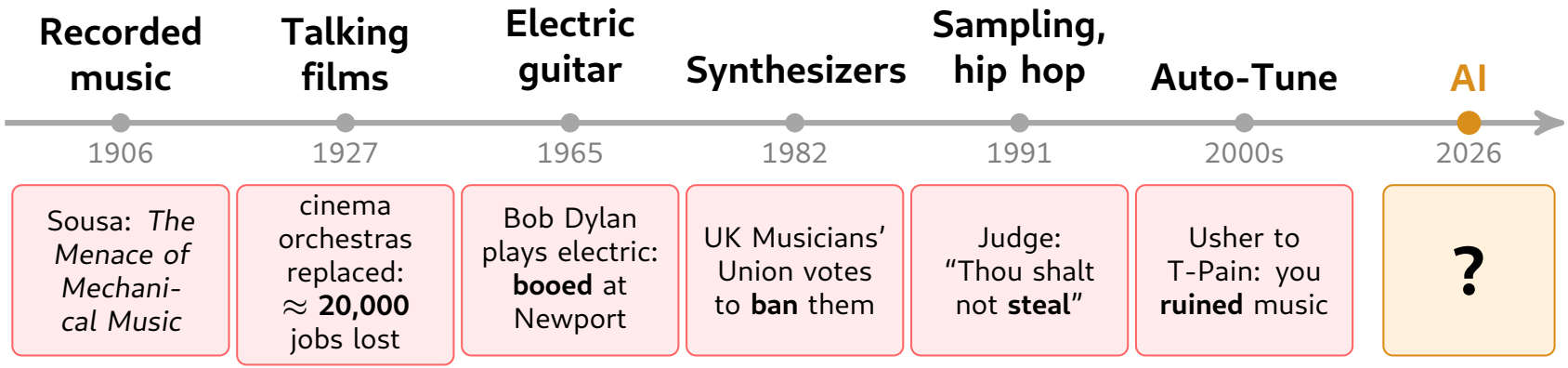
What would make AI an instrument?

Data: licensed or free of rights (Stable Audio Open; *Fairly Trained* label).
Control: the musician keeps the choices. A prompt alone does not make you the author (US Copyright Office, 2025).

The technologies that lasted found regulations and left the choices to musicians. AI has neither, yet.

Sousa, *Appleton's Magazine*, 1906 J. P. Kraft, *Stage to Studio*, 1996 Grand Upright Music v. Warner Bros., S.D.N.Y. 1991
T-Pain, *This Is Pop* (Netflix), 2021 Stable Audio Open, `https://arxiv.org/abs/2407.14358`
`https://www.fairlytrained.org` US Copyright Office, Jan 2025

Is AI the problem, or the next instrument?



first seen as a threat to music ⟶ all mainstream today

but "mainstream" does not mean "nobody lost": some fears were right

The sampling precedent

After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

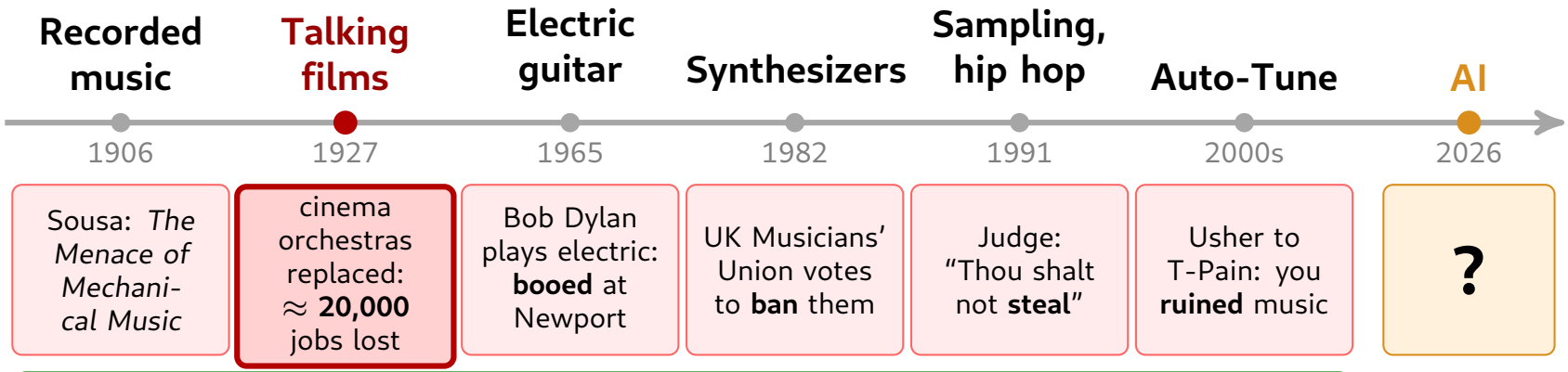
What would make AI an instrument?

Data: licensed or free of rights (Stable Audio Open; *Fairly Trained* label).
Control: the musician keeps the choices. A prompt alone does not make you the author (US Copyright Office, 2025).

The technologies that lasted found regulations and left the choices to musicians. AI has neither, yet.

Sousa, *Appleton's Magazine*, 1906 J. P. Kraft, *Stage to Studio*, 1996 Grand Upright Music v. Warner Bros., S.D.N.Y. 1991
T-Pain, *This Is Pop* (Netflix), 2021 Stable Audio Open, `https://arxiv.org/abs/2407.14358`
`https://www.fairlytrained.org` US Copyright Office, Jan 2025

Is AI the problem, or the next instrument?



first seen as a threat to music ⟶ all mainstream today

but "mainstream" does not mean "nobody lost": some fears were right

The sampling precedent

After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

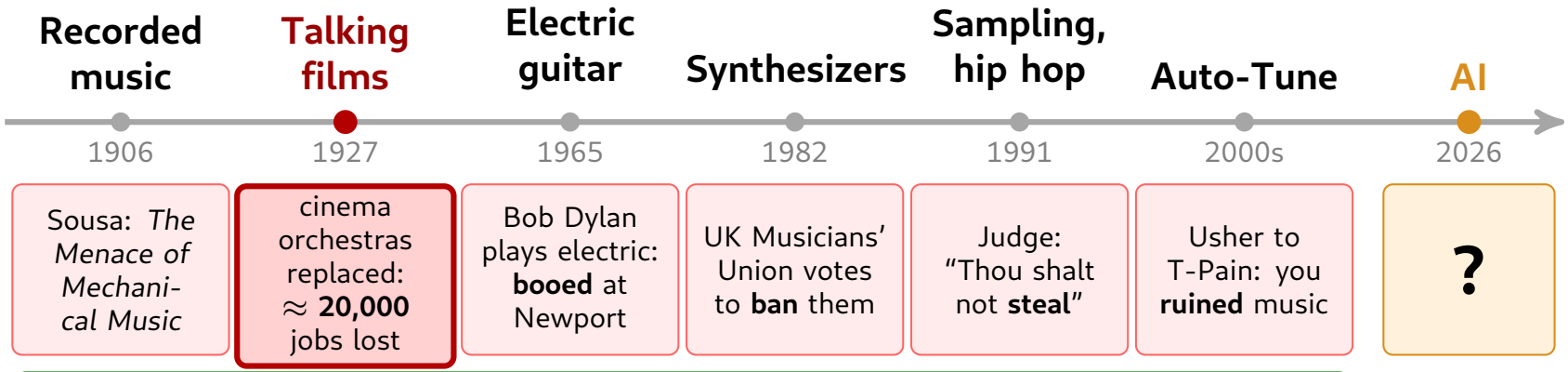
What would make AI an instrument?

Data: licensed or free of rights (Stable Audio Open; *Fairly Trained* label).
Control: the musician keeps the choices. A prompt alone does not make you the author (US Copyright Office, 2025).

The technologies that lasted found regulations and left the choices to musicians. AI has neither, yet.

Sousa, *Appleton's Magazine*, 1906 J. P. Kraft, *Stage to Studio*, 1996 Grand Upright Music v. Warner Bros., S.D.N.Y. 1991
T-Pain, *This Is Pop* (Netflix), 2021 Stable Audio Open, `https://arxiv.org/abs/2407.14358`
`https://www.fairlytrained.org` US Copyright Office, Jan 2025

Is AI the problem, or the next instrument?



first seen as a threat to music ⟶ all mainstream today
but "mainstream" does not mean "nobody lost": some fears were right

The sampling precedent

After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

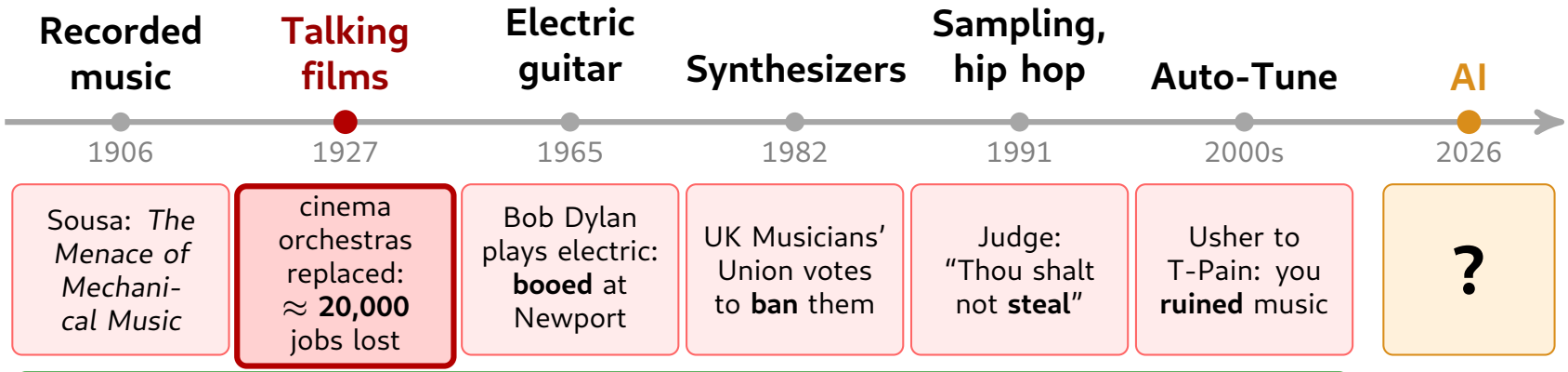
What would make AI an instrument?

Data: licensed or free of rights (Stable Audio Open; *Fairly Trained* label).
Control: the musician keeps the choices. A prompt alone does not make you the author (US Copyright Office, 2025).

The technologies that lasted found regulations and left the choices to musicians. AI has neither, yet.

Sousa, *Appleton's Magazine*, 1906 J. P. Kraft, *Stage to Studio*, 1996 Grand Upright Music v. Warner Bros., S.D.N.Y. 1991
T-Pain, *This Is Pop* (Netflix), 2021 Stable Audio Open, `https://arxiv.org/abs/2407.14358`
`https://www.fairlytrained.org` US Copyright Office, Jan 2025

Is AI the problem, or the next instrument?



first seen as a threat to music ⟶ all mainstream today
but "mainstream" does not mean "nobody lost": some fears were right

The sampling precedent

After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

What would make AI an instrument?

Data: licensed or free of rights (Stable Audio Open; *Fairly Trained* label).
Control: the musician keeps the choices. A prompt alone does not make you the author (US Copyright Office, 2025).

The technologies that lasted found regulations and left the choices to musicians. AI has neither, yet.

Sousa, *Appleton's Magazine*, 1906 J. P. Kraft, *Stage to Studio*, 1996 Grand Upright Music v. Warner Bros., S.D.N.Y. 1991
T-Pain, *This Is Pop* (Netflix), 2021 Stable Audio Open, `https://arxiv.org/abs/2407.14358`
`https://www.fairlytrained.org` US Copyright Office, Jan 2025

Place yourself: Generative AI: Instrument, or parasite of the artists?

pick one side

INSTRUMENT *or* **PARASITE**

Strongest for "instrument"

- Access: **2M** paying users, and tools for musicians,
- It restores and **gives voices back**,
- Every music technology was first a **threat**; sampling got rules,
- Humans learn from music too; UK court: weights hold **patterns**, not works.

Strongest for "parasite"

- "Essentially all music" as data, **no consent**, in a black box,
- Munich: **memorisation**, GEMA wins against Suno,
- Money to labels; **24%** of revenues at risk (projection),
- **52%** of uploads, up to **85%** of their streams bots.

Getty Images v. Stability AI, [2025] EWHC 2863 (Ch) GEMA v. Suno, LG München I, Jul 2026 CISAC and PMP Strategy, Dec 2024 other sources: previous slides

Outline

1 Agentic AI: Will AI destroy jobs?

2 Environment or sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate

Two sessions: prepare, then debate

1. **Work.** Will AI destroy jobs?
   yes ↔ no
2. **Environment or sovereignty.** Should France build its own frontier model?
   yes ↔ no
3. **Artistic expression.** Generative AI: Instrument, or parasite of the artists?
   instrument ↔ parasite

Session 1: become an expert

- You prepare **one** question, **on your own**,
- Read the slides, their notes, **and their sources**; look for new ones,
- Prepare arguments for **both** sides,
- On your question, you have **priority** at the start of the debate to introduce the first arguments.

Session 2: the moving debate

- **20 minutes** per question,
- The room is split in two: one side per answer,
- Experts open, then everyone joins,
- You **move** when an argument convinces you.

Two sessions: prepare, then debate

1. **Work.** Will AI destroy jobs?
   yes ↔ no
2. **Environment or sovereignty.** Should France build its own frontier model?
   yes ↔ no
3. **Artistic expression.** Generative AI: Instrument, or parasite of the artists?
   instrument ↔ parasite

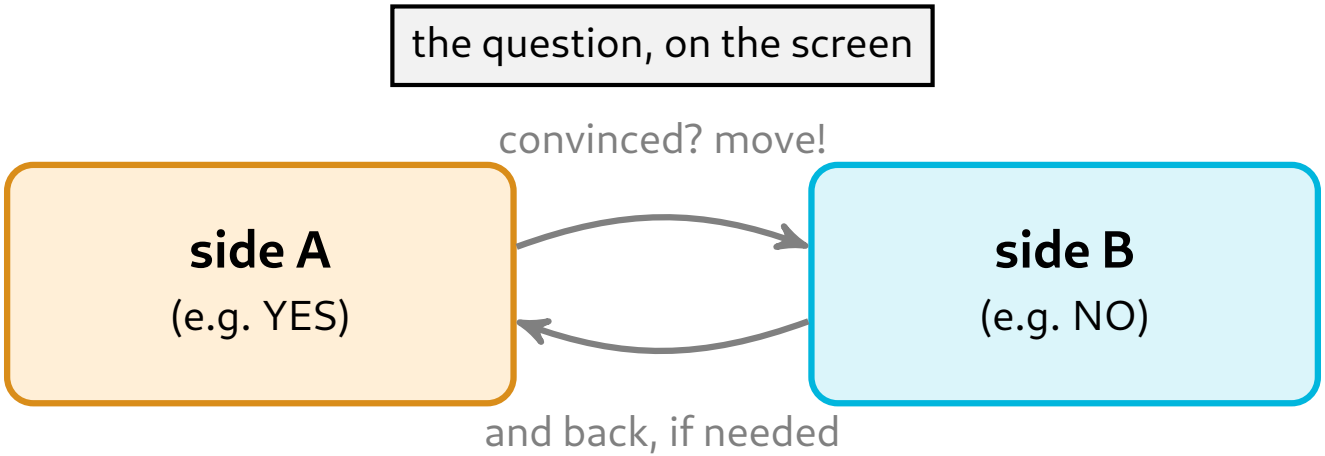
Session 1: become an expert

- You prepare **one** question, **on your own**,
- Read the slides, their notes, **and their sources**; look for new ones,
- Prepare arguments for **both** sides,
- On your question, you have **priority** at the start of the debate to introduce the first arguments.

Session 2: the moving debate

- **20 minutes** per question,
- The room is split in two: one side per answer,
- Experts open, then everyone joins,
- You **move** when an argument convinces you.

The rules of the moving debate



Speaking

- **Everyone speaks at least once**, during the session,
- **One side after the other**: an argument for A, then one for B,
- **Share the speaking time**: be short, leave room for others.

Listening

- **Listen** to the end, without interrupting,
- **Kindness**: criticise the argument, never the person,
- Answer the argument that was given, not another one.

It is a *moving* debate

Your position is where you stand. If an argument convinces you, **change sides**, physically. Changing your mind is not losing: it is the point of the exercise.